CHAPTER

33

Nose, nasal cavity and paranasal sinuses

Box 33.1 A note on terminology

'The advent of endoscopic sinus surgery led to a resurgence of interest in the detailed anatomy of the internal nose and paranasal sinuses. However, the official *Terminologia Anatomica* used by basic anatomists omits many of the structures of surgical importance. This led to numerous clinical anatomy papers and much discussion about the exact names and definitions for the structures of surgical relevance. The European position paper on the anatomical terminology of the internal nose and paranasal sinuses was conceived to re-evaluate the anatomical terms in common usage by endoscopic sinus surgeons and to compare this with the official *Terminologia Anatomica*' (Lund et al 2014).

Where appropriate, the terms used in this chapter are those commonly used in surgical practice as set out in the 'European position paper on the anatomical terminology of the internal nose and paranasal sinuses' (Table 33.1). Conflicts inevitably arise but hopefully have been minimized. The terms 'concha' and 'turbinate' are often used interchangeably but anatomists rarely use the latter term, while clinicians rarely use the former; in this chapter, concha is used to describe the bony structure seen in dried skulls, while turbinate is used to describe the bony structures, together with their overlying soft tissue and mucosa, that are encountered during surgery. 'Ethmoturbinal' does not appear in *Terminologia Anatomica* but is the collective term for the superior and middle turbinates, occasionally supplemented by a supreme turbinate. With regard to the terms 'skull base' and 'cranial base', although they are frequently used as though synonymous (the term 'skull base' is favoured in the position paper), the anterior skull base includes the inferior surface of the facial skeleton and will therefore be distinguished from the anterior cranial base, which is delimited by the inner and outer surfaces of the anterior cranial fossa (Ch. 27).

Box 33.1 and Table 33.1 present a note on terminology relating to the structures described in this chapter.

NOSE

The nose is the first part of the upper respiratory tract and is responsible for warming, humidifying and, to some extent, filtering inspired air. It also houses the olfactory epithelium, which contains olfactory receptor neurones responsible for detecting airborne odorant molecules.

The nose may be subdivided into an external nose, which opens anteriorly on to the face through the nostrils or nares, and an internal chamber, divided sagittally by a septum into right and left cavities, which open posteriorly into the nasopharynx through the posterior nasal apertures or choanae. The nasal cavities are housed in a supporting framework composed of bone and fibroelastic cartilages. The larger bones in this framework contain air-filled spaces lined with respiratory epithelium, described collectively as the paranasal sinuses. The sinuses and the nasolacrimal ducts drain into the nasal cavity via openings in its lateral walls (Fig. 33.1).

EXTERNAL NOSE

The external nose is a pyramidal structure located in the midline of the midface and attached to the facial skeleton. Its upper angle or root is

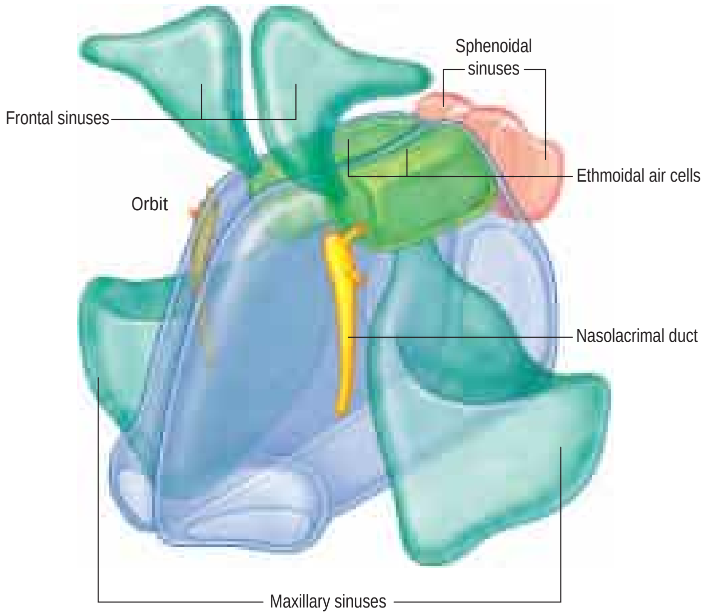

Fig. 33.1 An overview of the spatial relationships between the nasal cavity, paranasal sinuses and nasolacrimal ducts. (From Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

continuous with the forehead, and its free tip forms the apex, which projects anteriorly. The overall shape of the external nose is very variable. The lateral surfaces of the nose unite in the median plane to form the dorsum, which is narrowest at the medial canthus. The lobule is an area containing the tip of the nose. Its base contains two ellipsoidal apertures, the external nares or nostrils, which open on to its inferior surface, separated by the nasal septum and columella. The columella usually projects below the alar margin. The alar sulcus is a groove in the skin bounding the nasal alae above and joining the nasiolabial sulcus. Below, it curves towards the tip of the nose but does not reach it.

FACIAL AND NASAL PROPORTIONS

Available with the *Gray's Anatomy* e-book

SKIN AND SOFT TISSUE

The skin and soft tissue covering the nose vary in thickness. They are usually thin over the dorsum in the mid third, especially at the osseocartilaginous junction, the rhinion, and loosely connected to the nasal aponeurosis and the muscle fibres that fan out within it. It is thicker over the nasofrontal angle, and at the tip, where it has numerous large sebaceous glands and is more adherent. These variations affect the final nasal contour and profile after rhinoplasty.

The skin of the nose is separated from the underlying osteocartilaginous framework by four layers. These are the superficial fatty panniculus; the fibromuscular layer, a continuation of the facial superficial musculo-aponeurotic system (SMAS) (see chapter 30); the deep fatty layer; and the periosteum or perichondrium.

Table 33.1 Terminology

	Present 'surgical' terminology	Rhinological and anatomical synonyms (textbooks, literature)	*Terminologia Anatomica*	Suggested English terminology (position paper)	Frequency of variant in literature*
1	**Nasal cavity**	Inner nose Cavum nasi	Cavitas nasi	***Nasal cavity***	
1.1	Lateral nasal wall	Lateral nasal wall	n.e.	***Lateral nasal wall***	
1.2	Floor of nasal cavity	Nasal floor	n.e.	***Nasal floor***	
1.3	Nasal septum	Septum nasi	Septum nasi	***Nasal septum***	
1.3.1	Cartilaginous portion	Cartilaginous part of the nasal septum Cartilaginous segment Septal cartilage Lamina quadrangularis	Pars cartilaginea (septi nasi) Cartilago septi nasi	***Septal cartilage***	
1.3.2	Bony part	Bony/osseous septum Bony/osseous part of the nasal septum	Pars ossea septi nasi	***Bony septum***	
1.3.2.1	Lamina perpendicularis	Perpendicular plate of ethmoid	Lamina perpendicularis ossis ethmoidalis	***Perpendicular plate of ethmoid***	
1.3.2.2	Vomer	Vomer	Pars ossea septi nasi	***Vomer***	
1.3.3	Membranous portion	Membranous portion	Pars membranacea septi nasi	***Membranous portion (of nasal septum)***	
1.3.4	Jacobson's organ	Vomero-nasal organ	Organum vomeronasale	***Vomero-nasal organ***	
1.3.5	Septal tubercle	Tuberculum septi nasi Zuckerkandl's tubercle Morgagni's tubercle Septal swell body	n.e.	***Septal tubercle***	
1.4	Inferior turbinate	Inferior nasal turbinate Maxilloturbinal Concha inferior Lower turbinate	Concha nasi inferior	***Inferior turbinate***	
1.4.1	Inferior meatus	Inferior nasal meatus Lower nasal meatus	Meatus nasi inferior	***Inferior meatus***	
1.4.1.1	Nasolacrimal duct opening	Hasner's valve (Naso)lacrimal duct ostium Ostium lacrimale	Apertura/ostium ductus nasolacrimalis	***Nasolacrimal duct opening***	
1.5	Middle turbinate	Middle nasal turbinate First (persisting) ethmoturbinal First ethmoidal turbinate Middle concha Concha media	Concha nasi media	***Middle turbinate***	
1.5.1	Basal lamella of middle turbinate	Ground lamella of middle turbinate Third basal lamella	n.e.	***Basal lamella of middle turbinate***	
1.5.2	Paradoxically curved middle turbinate	Concave middle turbinate Inverse middle turbinate	n.e.	***Paradoxical middle turbinate***	3–26%
1.5.3	Concha bullosa (of middle turbinate)	Bullosa middle turbinate/concha	n.e.	***Concha bullosa (of middle turbinate)***	17–36% ~50% in Turkish
1.5.3.1	Interlamellar cell	Interlamellar cell	n.e.	***Interlamellar cell***	
1.6	Middle meatus	Meatus medius Middle nasal meatus	Meatus nasi medius	***Middle meatus***	
1.7	Ostiomeatal complex	Ostiomeatal complex	n.e.	***Ostiomeatal complex***	
1.8	Superior turbinate	Superior nasal turbinate Second (persisting) ethmoturbinal Second ethmoidal turbinate Superior concha Concha superior	Concha nasi superior	***Superior turbinate***	
1.8.1	Concha bullosa (of superior turbinate)	Concha bullosa (of superior turbinate)	n.e.	***Concha bullosa (of superior turbinate)***	1–2%
1.9	Superior meatus	Superior nasal meatus Upper nasal meatus	Meatus nasi superior	***Superior meatus***	
1.10	Supreme turbinate	Supreme nasal turbinate Third (persisting) ethmoturbinal Third ethmoidal turbinate Supreme concha Highest nasal concha Concha (nasalis) suprema (Morgagni)	Concha nasi suprema	***Supreme turbinate***	
1.11	Supreme meatus	Supreme nasal meatus	n.e.	***Supreme meatus***	
2	**Spheno-ethmoidal recess**	Recessus spheno-ethmoidalis	Recessus sphenoethmoidalis	***Spheno-ethmoidal recess***	
3	**Sphenopalatine foramen**	Foramen of sphenopalatine artery	Foramen sphenopalatinum	***Sphenopalatine foramen***	
4	**Olfactory cleft**	Olfactory ridge Olfactory groove Olfactory fissure Olfactory area	Sulcus olfactorius	***Olfactory cleft***	
4.1	Olfactory fibre(s)	Olfactory fibre(s) Fila olfactoria	Fila olfactoria (sing: filum olfactorium)	***Olfactory fibre(s)***	
5	**Choana (plur: choanae)**	Posterior nasal aperture(s) Nares posteriores	Choana (plur: choanae) Apertura nasalis posterior	***Choana***	
6	**Maxillary sinus**	Maxillary antrum	Sinus maxillaris	***Maxillary sinus***	
6.1	Maxillary sinus ostium	Maxillary opening	n.e.	***Maxillary sinus ostium***	
6.1.1	Accessory maxillary ostium (plur: ostia)	Additional maxillary sinus ostium	n.e.	***Accessory ostium***	5% normal 25% CRS patients
6.1.2	Maxillary hiatus	Maxillary hiatus	Hiatus maxillaris	***Maxillary hiatus***	
6.2	Infraorbital nerve canal	Infraorbital canal	Canalis infraorbitalis	***Infraorbital canal***	

Continued

CHAPTER 33

Table 33.1 **Terminology—cont'd**

	Present 'surgical' terminology	Rhinological and anatomical synonyms (textbooks, literature)	*Terminologia Anatomica*	Suggested English terminology (position paper)	Frequency of variant in literature*
6.3	Zygomatic recess	Recessus zygomaticus	n.e.	***Zygomatic recess***	
6.4	Alveolar recess	Recessus alveolaris	n.e.	***Alveolar recess***	
6.5	Prelacrimal recess	Prelacrimal recess	n.e.	***Prelacrimal recess***	
6.6	Lacrimal eminence	Eminentia lacrimalis Bulging of nasolacrimal duct	n.e.	***Lacrimal eminence***	
6.7	Canine fossa	Canine fossa Fossa canina	Fossa canina	***Canine fossa***	
6.8	Anterior (nasal) fontanelle	Fontanella nasal anterior	n.e.	***Anterior fontanelle***	
6.9	Posterior (nasal) fontanelle	Fontanella nasal posterior	n.e.	***Posterior fontanelle***	
6.10	Maxillary artery	(Internal) maxillary artery	Arteria maxillaris	***Maxillary artery***	
7	**Ethmoidal complex**	Ethmoid Ethmoidal sinus(es) Ethmoidal labyrinth Labyrinthus ethmoidalis	Cellulae ethmoidales	***Ethmoidal complex***	
7.1	Anterior ethmoidal cells	Anterior ethmoid Sinus ethmoidalis anterior Cells of anterior ethmoid Anterior ethmoid complex	Cellulae ethmoidales anteriores	***Anterior ethmoidal cells***	
7.2	Middle ethmoidal cells		Cellulae ethmoidales mediae	***t.b.a.***	
7.3	Posterior ethmoidal cells	Posterior ethmoid Sinus ethmoidalis posterior Dorsal ethmoidal cells Cells of posterior ethmoid	Cellulae ethmoidales posteriores	***Posterior ethmoidal cells***	
7.4	Anterior ethmoidal artery	Anterior ethmoidal artery	Arteria ethmoidalis anterior	***Anterior ethmoidal artery***	
7.5	Middle ethmoidal artery	Third ethmoidal artery Accessory ethmoidal artery Intermediate ethmoidal artery Arteria ethmoidalis tertia (40%)	n.e.	***Accessory ethmoidal artery***	(Var) up to 45% if it equates to any situation where >2 arteries
7.6	Posterior ethmoidal artery	Posterior ethmoidal artery	Arteria ethmoidalis posterior	***Posterior ethmoidal artery***	
8	**Anterior ethmoidal complex**	Anterior ethmoidal cells	Cellulae ethmoidales anteriores	***Anterior ethmoidal complex***	
8.1	Agger nasi	Operculum conchae mediae	Agger nasi	***Agger nasi***	
8.1.1	Agger nasi cell	Pneumatized agger nasi Agger cell	n.e. (cellula ethmoidalis anterior)	***Agger nasi cell***	>90%
9	**Uncinate process**	Uncinate process	Processus uncinatus	***Uncinate process***	
9.1	Deflected uncinate process	Doubled middle turbinate Anteriorly curved uncinate process Everted uncinate process	n.e.	***Everted uncinate process***	5–22%
9.2	Aerated uncinate process	Bullous uncinate process Pneumatized uncinate process	n.e.	***Aerated uncinate process***	1–2%
9.3	Basal lamella of uncinate process	Ground lamella of uncinate process Uncinate lamella First basal lamella	n.e.	***Basal lamella of uncinate process***	
9.4	Hiatus semilunaris	Semilunar hiatus Hiatus semilunaris inferior Semilunar gap	Hiatus semilunaris	***Inferior semilunar hiatus***	
9.4	Hiatus semilunaris (superior)	Hiatus semilunaris superior Hiatus semilunaris posterior Superior semilunar hiatus	n.e.	***Superior semilunar hiatus***	(Var)
9.5	Ethmoidal bulla	Bulla ethmoidalis	Bulla ethmoidalis	***Ethmoidal bulla***	
9.5.1	Non-pneumatized ethmoidal bulla	Torus bullaris	n.e.	***t.b.a.***	8%
9.5.2	Bulla lamella	Second ground lamella Basal lamella of ethmoidal bulla Second basal lamella	n.e.	***Basal lamella of ethmoidal bulla***	
9.5.3	Suprabullar recess	Sinus lateralis Suprabullar cell Recessus bullaris	n.e.	***Suprabullar recess***	71%
9.5.4	Retrobullar recess	Hiatus semilunaris superior	n.e.	***Retrobullar recess***	94%
9.5.5	Supraorbital recess	Supraorbital cell Supraorbital ethmoid cell Cellula orbitalis	n.e.	***Supraorbital recess***	(Var) 17%
9.5.6	Infraorbital cell	Haller cell Orbito-ethmoidal cell	n.e.	***Infraorbital cell***	4–15%
9.6	Ethmoidal infundibulum	Ethmoidal infundibulum	Infundibulum ethmoidale	***Ethmoidal infundibulum***	
9.6.1	Terminal recess	Terminal recess of ethmoidal infundibulum Recessus terminalis	n.e.	***Terminal recess***	(Var) 49–85%
9.7	Frontal recess	Recessus frontalis Frontal outflow tract	n.e.	***Frontal recess***	
9.7.1	Infundibular cells	Infundibular cells	n.e.	***Anterior ethmoidal cells***	(Var)
9.7.2	Lacrimal cells	Lacrimal cells	n.e.	***Anterior ethmoidal cells***	(Var) 33%
9.7.3	Nasofrontal duct	Frontal outflow tract Frontal recess	Ductus nasofrontalis	***t.b.a.***	
9.7.4	Maxillary crest	Lacrimal crest Maxillary line	n.e.	***Lacrimal bulge***	
9.7.5	Ethmoidal crest	Crista ethmoidalis Ethmoidal crest of the palatine bone	Crista ethmoidalis	***Ethmoidal crest***	

Table 33.1 **Terminology—cont'd**

	Present 'surgical' terminology	Rhinological and anatomical synonyms (textbooks, literature)	*Terminologia Anatomica*	Suggested English terminology (position paper)	Frequency of variant in literature*
9.7.6	Frontal sinus drainage pathway	Nasofrontal duct Frontal outflow tract Frontal recess	n.e.	***Frontal sinus drainage pathway***	
10	**Frontal sinus**	Frontal sinus	Sinus frontalis	***Frontal sinus***	
10.1	Interfrontal septum	Frontal sinus septum	Septum sinuum frontalium	***Frontal intersinus septum***	
10.2	Frontal sinus infundibulum	Frontal sinus infundibulum	n.e.	***Frontal sinus infundibulum***	
10.3	Intrafrontal cells	Frontal sinus cells Kuhn type 3/4 cells	Bullae frontales (sing: bulla frontalis)	***Frontoethmoidal cells***	(Var)
10.4	Intersinus septal cell	Intersinus septal cell	n.e.	***Intersinus septal cell***	
10.5	Frontal bulla	Frontal bulla	n.e. (cellula ethmoidalis anterior)	***t.b.a.***	(Var)
10.6	Frontal sinus ostium	Frontal ostium Opening of frontal sinus	Apertura sinus frontalis	***Frontal sinus opening***	
10.7	Frontal beak	Nasal beak Superior nasal spine	Spina frontalis (ossis frontalis) Spina nasalis interna	***Frontal beak***	
11	**Posterior ethmoidal complex**	Posterior ethmoidal cells	Cellulae ethmoidales posteriores	***Posterior ethmoidal complex***	
11.1	Onodi cell	Spheno-ethmoidal cell Gruenwald cell	n.e. (cellula ethmoidalis posterior)	***Sphenoethmoidal cell***	4–65% 8–14% Caucasians, 26–29% Asians
11.2	Basal lamella of superior turbinate	Fourth basal lamella	n.e.	***Basal lamella of superior turbinate***	
11.3	Lamina papyracea	Medial orbital wall Papyraceous lamina	Lamina orbitalis ossis ethmoidalis	***Lamina papyracea***	
11.4	Orbital apex	Orbital apex	n.e.	***Orbital apex***	
11.5	Anulus of Zinn	Common tendinous ring Common anular tendon	Anulus tendineus communis	***Anulus of Zinn***	
11.6	Ophthalmic artery	Ophthalmic artery	Arteria ophthalmica	***Ophthalmic artery***	
12	**Sphenoid sinus**	Sphenoid sinus	Sinus sphenoidalis	***Sphenoid sinus***	
12.1	Intersphenoidal septum	Intersphenoidal septum Sphenoid sinus septum	Septum sinuum sphenoidalium	***Sphenoid intersinus septum***	
12.2	Accessory sphenoidal septum (plur: septa)	Incomplete sphenoidal septations Partial sphenoidal septations Sphenoid sinus subseptations	n.e.	***Sphenoid septations***	(Var) 76%
12.3	Sphenoid sinus ostium	Sphenoid (sinus) ostium Sphenoid (sinus) opening Natural sphenoid ostium	Ostium (apertura) sinus sphenoidalis (plur: ostia sinuum sphenoidalium)	***Sphenoid sinus ostium***	
12.4	Planum sphenoidale	Sphenoid sinus roof Jugum sphenoidale Sphenoidal yoke	Jugum sphenoidale	***Planum sphenoidale***	
12.5	Sellar floor	Floor of sella Sellar bulge	n.e.	***Sellar floor***	
12.6	Vidian canal	Pterygoid canal Canalis nervi pterygoidei	Canalis pterygoideus	***Pterygoid (Vidian) canal***	
12.7	Foramen rotundum	Canalis rotundus Round foramen	Foramen rotundum	***Foramen rotundum***	
12.8	Lateral recess of sphenoid sinus	Lateral recess of sphenoid sinus	n.e.	***Lateral recess of sphenoid sinus***	(Var)
12.9	Optic tubercle	Optical nerve tubercle Prominentia nervi optici	Tuberculum nervi optici	***Optic nerve tubercle***	
12.9.1	Optic nerve canal	Eminentia nervi optici Optic nerve bulging Optic nerve canal contour	Canalis opticus	***Optic nerve canal***	(Var)
12.9.2	Carotid artery prominence	Prominentia canalis carotici	n.e.	***Carotid artery bulge***	(Var)
12.9.3	Optico-carotid recess	Carotid-optical recess Infraoptical recess	n.e.	***Optico-carotid recess***	(Var)
12.9.4	Sternberg's canal	Canalis craniopharyngicus lateralis	n.e.	***Lateral craniopharyngeal (Sternberg's) canal***	4% adults
13	**Sphenoidal rostrum**	Rostrum	Rostrum sphenoidale	***Sphenoid rostrum***	
14	**Vomerovaginal canal**	Vomerovaginal canal	Canalis vomerovaginalis	***Vomerovaginal canal***	
15	**Palatovaginal canal**	Palatovaginal canal	Canalis palatovaginalis	***Palatovaginal canal***	
16	**Skull base**	Cranial base Basicranium	Basis cranii	***Skull base***	
16.1	Inner skull base	Internal surface of cranial base	Basis cranii interna	***Inner skull base***	
17	**Anterior cranial fossa**	Anterior cranial fossa	Fossa cranii anterior	***Anterior cranial fossa***	
17.1	Olfactory fossa	Ethmoidal notch Fovea ethmoidalis	n.e.	***Olfactory fossa***	
17.2	Cribriform plate	Lamina cribrosa Roof of inner nose	Lamina cribrosa (ossis ethmoidalis)	***Cribriform plate***	
17.2.1	Cribriform foramina	Cribriform openings	Foramina cribrosa	***Cribriform foramina***	
17.2.2	Lateral lamella of cribriform plate	Lateral lamella of cribriform plate	n.e.	***Lateral lamella of cribriform plate***	
17.3	Ethmoidal roof	Foveae ethmoidales (ossis frontalis)	n.e.	***Ethmoidal roof***	
17.4	Crista galli	Crista galli	Crista galli	***Crista galli***	
17.4.1	Pneumatized crista galli	Pneumatized crista galli	n.e.	***Pneumatized crista galli***	13%
17.5	Foramen caecum	Foramen caecum	Foramen caecum	***Foramen caecum***	Open (Var: 1.4%)

Continued

CHAPTER 33

Table 33.1 Terminology—cont'd

	Present 'surgical' terminology	Rhinological and anatomical synonyms (textbooks, literature)	*Terminologia Anatomica*	Suggested English terminology (position paper)	Frequency of variant in literature*
18	**Middle cranial fossa**	Middle cranial fossa	Fossa cranii media	***Middle cranial fossa***	
18.1	Sella	Hypophysial fossa Pituitary fossa	Sella turcica	***Sella (turcica)***	
18.2	Sellar tubercle	Suprasellar notch	Tuberculum sellae	***Tuberculum sellae***	
18.3	Dorsum sellae	Dorsum sellae	Dorsum sellae	***Dorsum sellae***	
18.4	Anterior clinoid process	Anterior clinoid process	Processus clinoideus anterior (plur: processus clinoidei anteriores)	***Anterior clinoid process***	Pneumatized (Var: 16.5%)
18.5	Posterior clinoid process	Posterior clinoid process	Processus clinoideus posterior (plur: processus clinoidei posteriores)	***Posterior clinoid process***	
19	**Posterior cranial fossa**	Posterior cranial fossa	Fossa cranii posterior	***Posterior cranial fossa***	
19.1	Clivus	Clivus	Clivus	***Clivus***	

Abbreviations: n.e., non existent; sing., singular; plur., plural; t.b.a., to be abandoned.
**The frequency of specific variations in the anatomy varies considerably in the literature, which relates to the definitions used, the methodology utilized, i.e. anatomical dissection or imaging, whether the study included normal controls and/or patients with chronic rhinosinusitis (CRS), and the ethnicity of the subjects.*
(With permission from Lund VJ, Stammberger H, Fokkens WJ, et al 2014 European position paper on the anatomical terminology of the internal nose and paranasal sinuses. Rhinology 50:Supp 24:1–34.)

The proportions of the nose and face, both from in front and from the side, are of enormous significance to the rhinoplastic surgeon. Aesthetic proportions of the nose vary depending on sex, age, ethnicity and facial characteristics; however, ranges of normality are described to assist in aesthetic assessment (Akguner et al 1998). The female nose is slightly smaller and narrower than the male nose; it is often slightly concave in profile view, with a slightly obtuse nasolabial angle (increased tip rotation). In terms of overall proportion, the face may be divided into horizontal thirds and vertical fifths, with the nose occupying the middle section of each. The width of the nose is roughly 70% of the length; the width of the alar base is usually equal to the intercanthal distance. The height of the nose is defined by tip projection, where the proportion of the length of a line from the tip to the alar groove to the length of a line from nasion to alar groove is in the range of 0.55–0.60. The nasolabial angle, reflecting upward rotation of the nose from the upper lip, normally lies within a range of 105–120° in females and 90–105° in males. On basal view, the nose is roughly shaped as an equilateral triangle. The nares usually measure 1.5–2 cm anteroposteriorly and 0.5–1 cm transversely, and are narrower in front; they occupy approximately two-thirds of the height of the base. The midline columella, containing the caudal end of the nasal septum and the medial crura of the lower lateral cartilages, usually extends 3–5 mm below the nares on lateral views (Fig. 33.2).

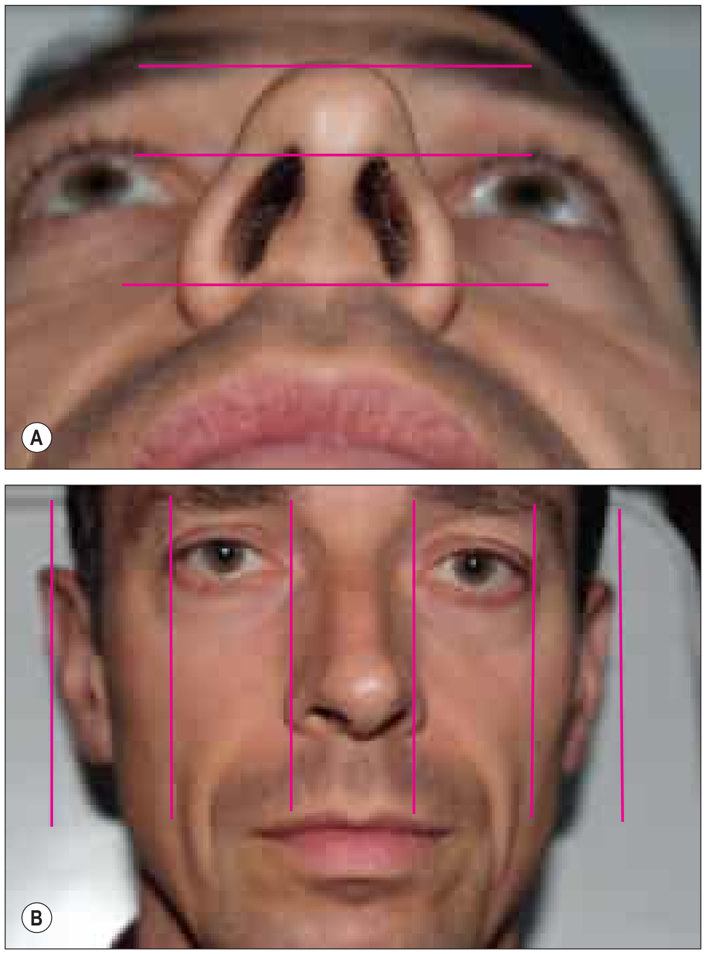

Fig. 33.2 The adult male nose. **A**, Basal view. **B**, Frontal view. The face can be divided into horizontal thirds and vertical fifths, with the nose filling the central segment in terms of both width and height. The basal view may also be divided into horizontal thirds, with the nostrils filling the lower two-thirds.

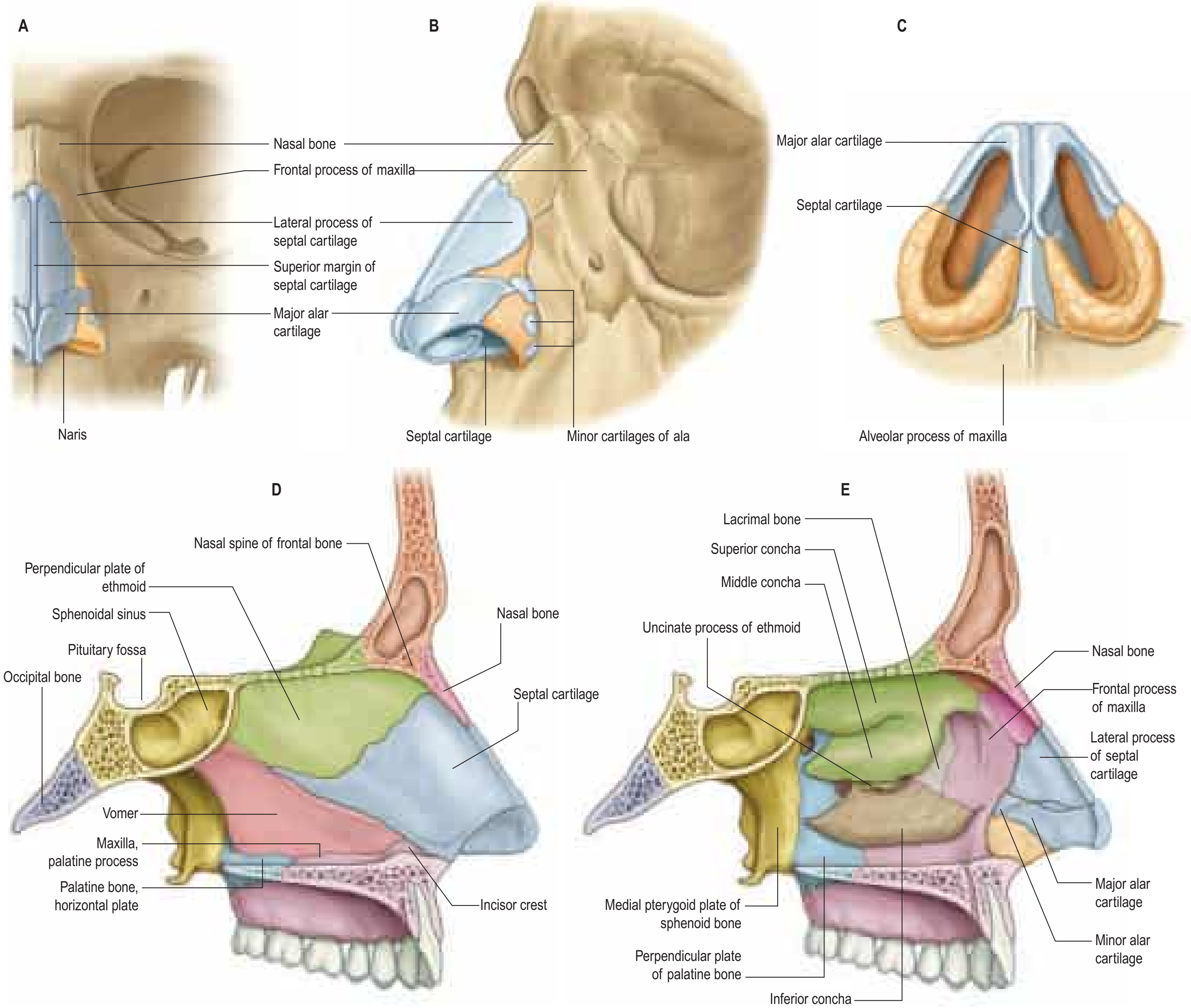

Fig. 33.3 The bony and cartilaginous skeletons of the nose. **A**, The external nose, frontal view. **B**, The external nose, lateral view. **C**, An inferior view of the cartilages. **D**, The nasal cavity, medial wall. **E**, The nasal cavity, lateral wall (left side).

Soft tissue areas of the nose

Four soft tissue areas of the nose lack cartilaginous support. They have been given numerous eponymous names and descriptions, but were reclassified by Huizing (2003).

Paraseptal soft tissue area

The supratip area contains the paired lateral cartilages, which gradually separate from the septum to a level just above the septal angle. The amount of flare varies, and encloses a small paraseptal soft tissue triangle on each side of the septum.

Lateral soft tissue area

The lateral margin of the lateral cartilage, the piriform aperture and the cranial margin of the lateral crus of the major alar cartilage enclose a triangle containing loose fibroareolar tissue, the transverse portion of nasalis, and one or more small sesamoid cartilages.

Caudal lobular notch

The domal segment of the intermediate crus of the major alar cartilage has a small indentation. The apex of the nostril is a soft tissue triangle with little soft tissue separating the internal and external skin. Incisions in this area may cause unsightly scarring and deformity.

Alar soft tissue area

The lateral crus fails to extend to the lateral limit of the lobule, forming the fourth soft tissue area.

BONES AND CARTILAGE

Bony skeleton of the external nose

The piriform aperture has sharp edges. It is bounded below and laterally by the maxilla and above by the nasal bones (Fig. 33.3). The lateral part of the inferior edge of the piriform aperture merges into its lateral wall, which is formed by the frontal process of the maxilla. It is bounded above by the nasal part of the frontal bone and superomedially by the lateral edge of the nasal bone.

The paired nasal bones vary in thickness and width, which is of significance in planning osteotomies. They are thickest and widest at the nasofrontal suture, narrow at the nasofrontal angle before they widen, and become thinner 9–12 mm below the nasofrontal angle. They average 25 mm in length but this can vary widely. The perpendicular plate of the ethmoid bone (part of the bony nasal septum) articulates with the undersurface of the nasal bones and provides support to the dorsum of the nose. A midline bony spine deep to the fused nasal bone projects inwards to articulate with the perpendicular plate of the ethmoid and fuses with the fibrous tissue connecting the lateral nasal cartilages and cartilaginous septum. This is known as the keystone area and provides essential support to the nasal dorsum.

Fractures of the nasal bones The most common injury to the facial skeleton is a fracture of the nasal bones. In simple fractures, the break often occurs between the proximal thicker bone and the thinner bone distally. Displaced fractures require reduction to avoid cosmetic deformity. The terminal branch of the anterior ethmoidal nerve and its accompanying vessels are at risk when injuries involve the dorsum of the nose.

Cartilaginous skeleton of the external nose

The cartilaginous framework consists of the paired lateral and major cartilages and several minor alar nasal cartilages (see Fig. 33.3).

Lateral (superior/upper lateral) nasal cartilage

The lateral nasal cartilage is triangular, and its anterior margin is thicker than the posterior margin. The upper part fuses with the septal cartilage, but anteroinferiorly, it may be separated from it by a narrow fissure. The superior margin of the lateral nasal cartilage is attached to the nasal bone and frontal process of the maxilla, and the inferior margin is connected by fibrous tissue to the lateral crus of the major alar cartilage. Laterally, the cartilage is attached indirectly to the margins of the piriform aperture by loose fibroareolar connective tissue, which may also contain one or more small sesamoid cartilages. The angle formed between the caudal end of the lower lateral and the septum, the internal nasal valve, is usually between 10 and 15° and represents the narrowest cross-sectional area and the area of greatest airflow resistance. Structural abnormalities in this area are likely to produce symptomatic nasal obstruction.

Major alar (lower lateral) cartilage

The major alar cartilage is a highly complex, thin, flexible plate, which is integral to the nasal lobule. It lies below the upper lateral cartilage and curves acutely around the anterior part of its naris. The medial part, the narrow medial crus (septal process), is loosely connected by fibrous tissue to its contralateral counterpart and to the anteroinferior part of the septal cartilage. The intermediate crus forms the margin of the apex of the nostril. The domes give rise to the tip-defining points of the nose. The lateral crus lies lateral to the naris and runs superolaterally away from the margin of the nasal ala. The upper border of the lateral crus of the major alar cartilage is attached by fibrous tissue to the lower border of the lateral nasal cartilage. Its lateral border is connected to the frontal process of the maxilla by a tough fibrous membrane containing three or four minor alar cartilages. The junction between the lateral crura of the major alar and lateral cartilages is variable; the two edges may form a ‘scroll’, with an outcurving of the lateral cartilage meeting an incurving of the major alar cartilage, in which case the lateral crus is then the more lateral at the junction. The lateral crus is shorter than the lateral margin of the naris; the most lateral part of the margin of the ala nasi is fibroadipose tissue covered by skin. In front, the angulations or ‘domes’ between the medial and lateral crurae of the major alar cartilages are separated by a notch palpable at the tip of the nose.

Alar cartilage morphology

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MUSCLES

The nasal muscle group includes procerus, nasalis, dilator naris anterior, depressor septi and levator labii superioris alaeque nasi (**Fig. 33.4**, see Figs 30.17, 30.18). These muscles are involved in respiration and facial expression. Any or all of these muscles may be absent in cleft lip deformities with corresponding functional and aesthetic consequences.

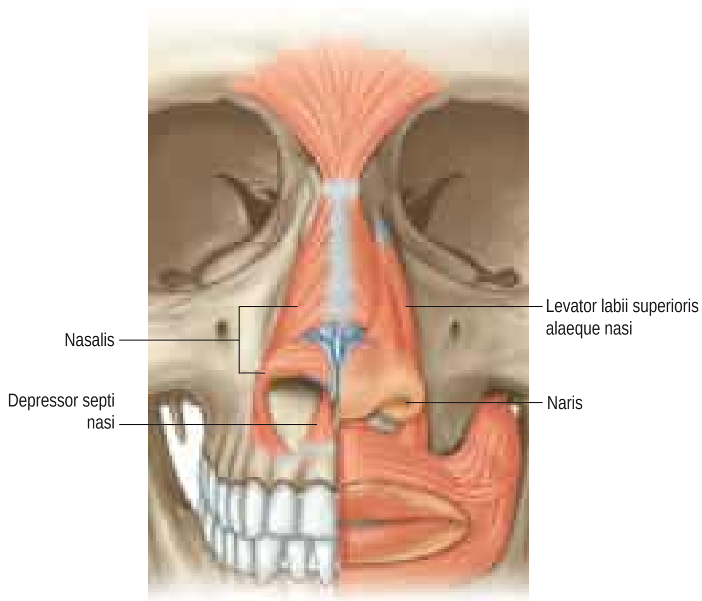

Fig. 33.4 The nasal musculature. (From Drake RL, Vogl AW, Mitchell A (eds), Gray’s Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

Procerus

Procerus is a small pyramidal muscle that lies close to, and is often partially blended with, the medial side of the frontal part of occipitofrontalis. It arises from a fascial aponeurosis attached to the periosteum covering the lower part of the nasal bone, the perichondrium covering the upper part of the lateral nasal cartilage, and the aponeurosis of the transverse part of nasalis. It is inserted into the glabellar skin over the lower part of the forehead between the eyebrows.

Vascular supply Procerus is supplied mainly by branches from the facial artery.

Innervation Procerus is supplied by temporal and lower zygomatic branches from the facial nerve (a supply from the buccal branch has been described).

Actions Procerus draws down the medial angle of the eyebrow and produces transverse wrinkles over the bridge of the nose. It is active in frowning and ‘concentration’, and helps to reduce the glare of bright sunlight. Not surprisingly, it is a common target in non-surgical facial rejuvenation techniques, using botulinum toxin.

Nasalis

Nasalis consists of transverse and alar components. The transverse part (compressor naris) is attached to the maxilla above and lateral to the incisive fossa, and lateral to the alar part. Its fibres pass upwards and medially, and expand into a thin aponeurosis that merges with its counterpart across the bridge of the nose, with the aponeuroses of procerus, and with fibres from levator labii superioris alaeque nasi. Fibres from the transverse part may also blend with the skin of the nasolabial and alar folds. The alar part (pars alaris or dilator naris posterior) is attached to the maxilla above the lateral incisor and canine, lateral to the bony attachment of depressor septi, and medial to the transverse part, with which it partly merges. Its fibres pass upwards and anteriorly, and are attached to the skin of the ala above the lateral crus of the lower lateral cartilage, and to the posterior part of the mobile septum. The pars alaris helps to produce the upper ridge of the philtrum.

Dilator naris anterior (also known as apicis nasi or the small dilator muscle of the nose) is a very small muscle attached to the upper lateral cartilage, the alar part of nasalis, the caudal margin of the lateral crus and the lateral alar crus. It encircles the naris and acts as a primary dilator of the nostril.

Vascular supply Nasalis is supplied by branches from the facial artery and from the infraorbital branch of the maxillary artery.

Innervation Nasalis is supplied by the buccal branch of the facial nerve. It may also be supplied by the zygomatic branch of the facial nerve.

Actions The transverse part compresses the nasal aperture at the junction of the vestibule and the nasal cavity. The alar parts draw the alae and posterior part of the columella downwards and laterally, and so assist in widening the nares and in elongating the nose. They are active immediately before inspiration. Dilator naris anterior and the alar part of nasalis (dilator naris posterior) probably function to prevent collapse of the nasal valve during inspiration. Their electromyographic activity is directly proportional to ventilatory resistance and is modified by signals that travel from pulmonary mechano- and pressure receptors via afferent vagal pathways to the brainstem respiratory centre; the efferent limb of the reflex arc runs in the facial nerve.

Depressor septi

Depressor septi lies immediately deep to the mucous membrane of the upper lip. It is usually attached to the periosteum covering the maxilla

The medial crus has two components, a footplate segment and a columellar segment, which angulate with each other in two planes; they diverge in the basal plane, and rotate upwards in the lateral plane. There is usually asymmetry in the pairs of medial crura.

The medial crus joins the intermediate crus at what is usually the most convex point of the columella, known as the columellar breakpoint.

The intermediate crus is also described in two components. The lobular segment is usually flared and forms the transition between the medial crus and the domal segment of the intermediate crus. The domal segment may be convex, producing an aesthetically pleasing tip; flat, giving a 'boxy' appearance; or convex, producing a 'double-dome'. The domal or tip-defining points are usually formed by the most anterior projection of the domal segment. The amount of divergence of the domes, and the thickness of the overlying soft tissue envelope, determine the relative position of the tip-defining points. The dome projects up to 8–10 mm caudal, and 3–6 mm anterior, to the anterior septal angle, the difference between the two creating the supratip break-point. Disruption of this relationship with rhinoplasty, with loss of projection of the tip, may produce a 'polybeak' deformity.

Classically, transverse connective tissue fibres have been described binding the medial and intermediate crura; interdomal, intercrural and septocrural ligaments have been described. Cadaveric studies by Zhai et al (1995) disputed the presence of transverse fibres, and found that all connective tissue fibres run parallel to the cartilages. These findings notwithstanding, the fibrous connections along the length of the medial and intermediate crura form a single functional unit in the tip.

The large lateral crus determines the shape of the alar lateral wall. Medially, it is a continuation of the intermediate crus, while laterally it connects with accessory cartilages. It runs at the caudal edge of the alar rim in the anterior half, then moves cephalically, leaving a soft tissue area in the rim laterally. Typically, the longitudinal axis of the lateral crus forms an angle of 45° with the septum. More vertical, or cephalic, positioning results in a 'parenthesis' tip deformity.

The lateral crus may take a highly variable position, being either convex or concave, or a combination of both, in medial and lateral portions; asymmetry from side to side has been reported in over half of anatomical specimens.

A chain of lateral accessory cartilages with dense fibrous attachments connect to the lateral crus and the piriform aperture, and to the anterior nasal spine through connection in the floor of the nose.

Tip support

The inherent strength and shape of the cartilaginous framework, and its attachments to surrounding structures, provide support to the tip of the nose. Typically, 'major' and 'minor' tip support mechanisms are described.

Major tip support mechanisms The major mechanisms supporting the tip are: the size, shape and strength of the major alar cartilages; the medial crural footplate attachment to the caudal part of the septum; the attachment of the caudal border of the lateral cartilages to the cephalic border of the major alar cartilages; and the cartilaginous dorsal septum.

Minor tip support mechanisms The minor mechanisms supporting the tip are: the ligamentous sling spanning the domes of the lower lateral cartilages (i.e. the interdomal ligament); the sesamoid complex of major alar cartilages; the attachment of the major alar cartilages to the overlying skin/soft tissue envelope; the nasal spine; and the membranous septum.

Rhinoplasty approaches, either through an intercartilaginous incision between the lateral and the major alar cartilages (closed rhinoplasty), or through an incision caudal to the major alar cartilages and degloving the entire cartilaginous framework (open rhinoplasty), disrupt the tip support mechanisms; the integrity of these mechanisms must be restored during the procedure to prevent loss of tip support and subsequent tip ptosis.

above the central and lateral incisors and the anterior nasal spine, and to the fibres of orbicularis oris above the central incisor. Its fibres pass to the columella, the mobile part of the nasal septum and the base of the medial crus of the nasal cartilage. A few muscle slips may pass between the medial crura into the nasal tip. Depressor septi may be absent or rudimentary.

Vascular supply Depressor septi is supplied by the superior labial branch of the facial artery.

Innervation Depressor septi is innervated by the buccal branch, and sometimes by the zygomatic branch, of the facial nerve.

Actions Depressor septi pulls the columella, the tip of the nose and the nasal septum downwards. It tenses the nasal septum at the start of nasal inspiration and, with the alar part of nasalis, widens the nasal aperture, as well as causing the nose to 'dip' when some people smile.

Levator labii superioris alaeque nasi

Levator labii superioris alaeque nasi arises from the upper part of the frontal process of the maxilla and, passing obliquely downwards and laterally, divides into medial and lateral slips. The medial slip blends into the perichondrium of the lateral crus of the major alar cartilage of the nose and the skin over it. The lateral slip is prolonged into the lateral part of the upper lip, where it blends with levator labii superioris and orbicularis oris. Superficial fibres of the lateral slip curve laterally across the front of levator labii superioris and attach along the floor of the dermis at the upper part of the nasolabial furrow and ridge.

Vascular supply Levator labii superioris alaeque nasi is supplied by the facial artery and the infraorbital branch of the maxillary artery.

Innervation Levator labii superioris alaeque nasi is innervated by zygomatic and superior buccal branches of the facial nerve.

Actions The lateral slip raises and everts the upper lip and raises, deepens and increases the curvature of the top of the nasolabial furrow. The medial slip pulls the lateral crus superiorly, displaces the circumalar furrow laterally, and modifies its curvature; it is a dilator of the naris. Depressor septi and the medial slip of levator labii superioris alaeque nasi are sometimes described as secondary nasal dilators; there is little evidence that either of these muscles has any direct influence on the nasal valve.

Anomalous nasal muscles

Anomalous, inconstant, nasal muscles have been described. They include anomalous nasi, attached to the frontal process of the maxilla, procerus, transverse part of nasalis and the upper lateral cartilage (i.e. in a region normally devoid of muscle), and compressor narium minor, which passes between the anterior part of the lower lateral cartilage and the skin near the margins of the nares. The existence of a small levator septi nasi has been questioned.

Superficial muscular aponeurotic system (SMAS)

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CUTANEOUS VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Nasal skin receives its blood supply from branches of the facial, ophthalmic and infraorbital arteries. The alae and lower part of the nasal septum are supplied by lateral nasal and septal branches of the facial artery, and the lateral aspects and dorsum of the nose are supplied by the dorsal nasal branch of the ophthalmic artery and the infraorbital branch of the maxillary artery. The venous networks draining the external nose do not run parallel to the arteries but correspond to arteriovenous territories of the face. The frontomedian region of the face, including the nose, drains to the facial vein, and the orbitopalpebral area of the face, including the root of the nose, drains to the ophthalmic veins. The connections of the veins of the nose, upper lip and cheek (the 'danger triangle of the face') with the drainage area of the valveless ophthalmic veins, and hence to the cavernous sinus, are clinically significant because they can be a route for spreading infection that initiates thrombosis of the major intracranial sinuses. Lymph drainage is primarily to the submandibular group of nodes, although lymph draining from the root of the nose drains to superficial parotid nodes.

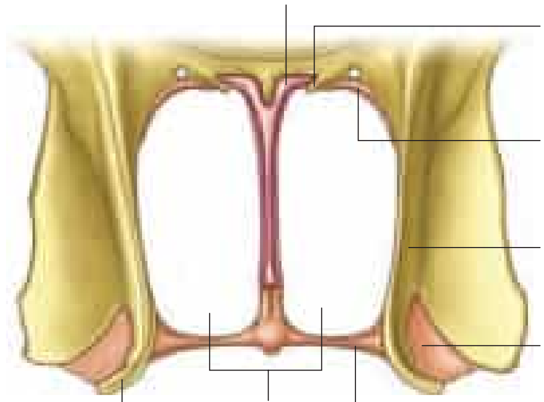

Fig. 33.5 A posterior view of the choanae.

CUTANEOUS INNERVATION

Nasal skin is innervated by the infratrochlear and external nasal branches of the nasociliary nerve (ophthalmic division, trigeminal nerve), and the nasal branch of the infraorbital nerve (maxillary division, trigeminal nerve).

NASAL CAVITY

The nasal cavity is an irregular space between the roof of the mouth and the cranial base. It is wider below than above, and widest and vertically deepest in its central region, where it is divided by a vertical, midline, osseocartilaginous septum. The bony part of the septum reaches the posterior limit of the cavity.

The nasal cavity communicates with the paranasal sinuses and opens into the nasopharynx through a pair of oval openings, the posterior nasal apertures or choanae. The latter are separated by the posterior border of the vomer, and each is limited above by the vaginal process of the medial pterygoid plates, laterally by the perpendicular plate of the palatine bone and the medial pterygoid plate, and below by the horizontal plate of the palatine bone (Fig. 33.5). The parameters of the intranasal spaces depend on age and gender: growth is usually completed by the age of 16 (Samoliński et al 2007). The adult choana typically measures 2.5 cm in vertical height and 1.3 cm transversely; size is not usually affected by deviations of the nasal septum. The vomerovaginal and palatovaginal canals are found in the roof of this region.

Each half of the nasal cavity has a vestibule, roof, floor, medial (septal) and lateral walls.

NASAL VESTIBULE

The nasal vestibule lies just inside the naris. It is limited above and behind by a curved ridge, the limen nasi, raised where the greater ala of the lateral cartilaginous crus overlaps the lower edge of the lateral nasal cartilage on each side. On the septal side of the nasal cavity, the superior edge of the medial crus of the major alar cartilage (the medial intumescence) marks the boundary between the nasal vestibule and the nasal cavity. The medial wall of the vestibule is formed by a mobile septum consisting of the columella (which does not contain cartilage) and the underlying medial crura of the alar cartilages.

ROOF

The roof is horizontal in its central part and slopes downwards in front and behind (see Fig. 33.3D and E). The anterior slope is formed by the nasal spine of the frontal bones and by the nasal bones. The central region is formed by the cribriform plate of the ethmoid bone, which separates the nasal cavity from the floor of the anterior cranial fossa. It contains numerous small perforations that transmit the olfactory nerves and their ensheathing meningeal layers, and a separate anterior foramen that transmits the anterior ethmoidal nerve and vessels. The height of the cranial base is greatest anteriorly; hence, when dissecting along the cranial base during sinus surgery, it is safest to do so from back to front, addressing the lower-lying posterior region first to avoid inadvertent

Although each muscle is independent in terms of innervation and function, the mimetic muscles of the nose form a continuous layer with connections between all the muscular and ligamentous components (Saban et al 2008). Thus, the superficial muscular aponeurotic system is continuous from the nasofrontal process to the nasal tip, splitting at the caudal end of the lateral cartilage into superficial and deep layers, each with medial and lateral components (Oneal et al 1999). Dissection in rhinoplasty is usually performed in a sub-superficial muscular aponeurotic system plane. (For further information on the superficial muscular aponeurotic system, see p. 476.)

intracranial penetration. Posteriorly, the roof of the nasal cavity is formed by the anterior aspect of the body of the sphenoid, interrupted on each side by an opening of a sphenoidal sinus, and the sphenoidal conchae or superior conchae.

FLOOR

The floor of the nasal cavity is smooth and concave transversely, and slopes up from the anterior to the posterior apertures. The greater part is formed by the palatine processes of the maxillae, which articulate posteriorly with the horizontal plates of the palatine bones at the palatomaxillary suture (see Figs 33.3D, E). Anteriorly, near the septum, a small infundibular opening in the bone of the nasal floor leads into the incisive canals that descend to the incisive fossa; this opening is marked by a slight depression in the overlying mucosa.

The floor of the nose may be deficient as a result of congenital clefting of the hard and/or soft palate.

MEDIAL WALL

The medial wall of each nasal cavity is the nasal septum, a thin sheet of bone (posteriorly) and cartilage (anteriorly) that lies between the roof and floor of the cavity (see Figs 30.4, 33.3D).

Bony septum

The septum is usually relatively featureless but sometimes exhibits ridges or bony spurs. The posterosuperior part of the septum and its posterior border are formed by the vomer, which extends from the body of the sphenoid to the nasal crest of the palatine bones and maxilla (see Fig. 30.4). The nasopalatine nerves and vessels groove its surface. (The nasopalatine artery, also known as the septal artery, is a branch of the maxillary artery; it leaves the pterygopalatine fossa through a canal inside the palatine bone, runs parallel to the nasopalatine nerve and ends in the incisive canal, where it anastomoses with the greater palatine artery.) The anterosuperior part of the septum is formed by the perpendicular plate of the ethmoid, which is continuous above with the cribriform plate and the frontal bone. Other bones that make minor contributions to the septum at the upper and lower limits of the medial wall are the nasal bones and the nasal spine of the frontal bones (anterosuperior), the rostrum and crest of the sphenoid (posterosuperior), and the nasal crests of the maxilla and palatine bones (inferior).

Cartilaginous septum

The septal cartilage is almost quadrilateral and may extend back (especially in children) for some distance between the vomer and the perpendicular plate of the ethmoid. Its anterosuperior margin is connected above to the posterior border of the internasal suture, and the distal end of its superior portion is continuous with the upper lateral cartilages. The anteroinferior border is connected by fibrous tissue on each side to the medial crurae of the major alar cartilage. Anteroinferiorly, the cartilaginous septum is attached to the anterior nasal spine, which is formed by anterior projections of each maxillary crest, and it has a strong, tongue-in-groove attachment with the premaxilla and vomer. The cartilaginous septum anterior to the spine is essential in tip support and should not be excised during septal surgery in order to prevent columellar retraction or loss of tip support. The posterosuperior border joins the perpendicular plate of the ethmoid, while the posteroinferior border is attached to the vomer and, anterior to that, to the nasal crest and anterior nasal spine of the maxilla. The anteroinferior part of the nasal septum between the nares is devoid of cartilage and is therefore called the membranous septum; it is continuous with the columella anteriorly.

Above the incisive canals, at the lower edge of the septal cartilage, a depression pointing downwards and forwards is all that remains of the nasopalatine canal, which connected the nasal and buccal cavities in early fetal life. Near this recess, a minute orifice leads back into a blind tubule, 2–6 mm long, which lies on each side of the septum and houses remnants of the vomeronasal organ (see below).

LATERAL WALL

The lateral wall of the nasal cavity is formed anteroinferiorly by the maxilla and its anterior and posterior fontanelles (bony deficiencies in

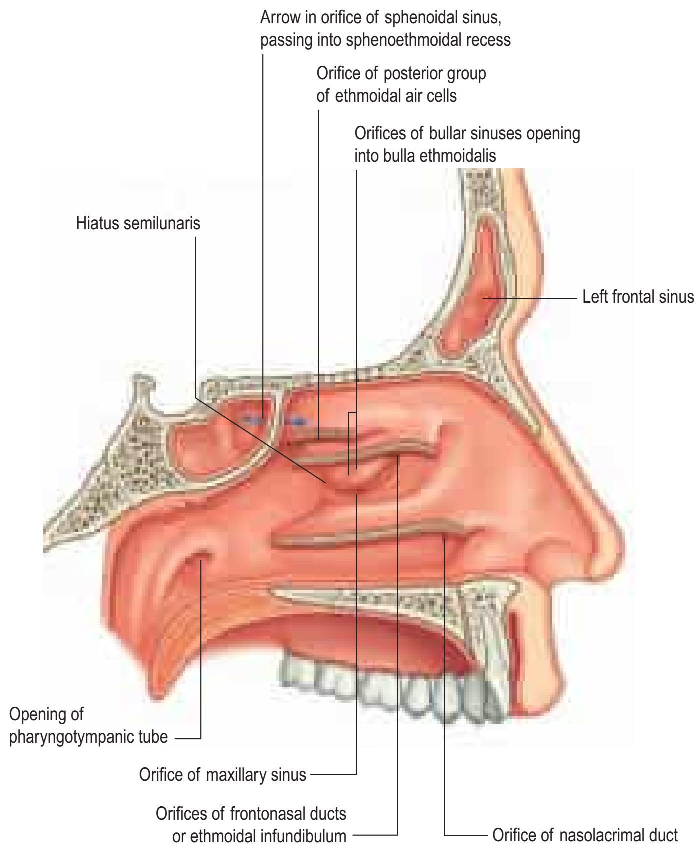

Fig. 33.6 The lateral wall of the nasal cavity. The conchae have been removed to show the positions of the ostia of the paranasal sinuses and the nasolacrimal duct. (From Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

the medial wall of the maxilla that are obliterated to varying degrees by fibrous tissue); posteriorly by the perpendicular plate of the palatine bone; and superiorly by the labyrinth of the ethmoid bone (**Fig. 33.6**; see Figs 33.3E, 30.5). It contains three projections of variable size: the inferior, middle and superior nasal conchae or turbinates.

The conchae curve inferomedially in general, each roofing a groove, or meatus, which is open to the nasal cavity. The middle concha may also curve inferolaterally; less commonly, it may sometimes be expanded by an enclosed air cell to form a so-called 'concha bullosa', or occasionally may have a concave medial surface, known as a paradoxical turbinate. The main features of the lateral nasal wall are a rounded elevation, the bulla ethmoidalis, and a curved cleft, the hiatus semilunaris, formed by the posterior edge of the uncinate process and the anterior face of the ethmoidal bulla. This constitutes the medial limit of the ethmoidal infundibulum, a slit-like space that leads towards the maxillary ostium. The maxillary ostium is normally found lateral to the anteroinferior aspect of the uncinate process. The latter may be attached to either the lateral nasal wall (50%), or the anterior cranial fossa (25%) or the middle concha (25%). Where the uncinate process is attached determines whether the frontal sinus drains lateral to the ethmoidal infundibulum or into it. If the uncinate process is attached to the lateral wall, the frontal sinus will drain into the middle meatus and not into the ethmoidal infundibulum, whereas with the other configurations, the sinus will drain into the infundibulum, and thus near or into the maxillary ostium. Agger nasi air cells are anterior ethmoidal air cells that lie anterior to the ethmoidal bulla (see Fig. 33.6). The posterior fontanelle lies posterior to the uncinate process where there is no bone in the medial wall of the maxillary sinus, inferoposterior to the hiatus, and frequently has an accessory opening (**Fig. 33.7**).

Inferior concha and inferior meatus

The inferior concha is a thin, curved, independent bone (for more details, see p. 483). It articulates with the nasal surface of the maxilla and the perpendicular plate of the palatine bone. Its free lower border is gently curved and the subjacent inferior meatus reaches the nasal floor. The inferior meatus is the largest meatus, extending along almost all the lateral nasal wall. It is deepest at the junction of its anterior and middle thirds, where it admits the inferior opening of the nasolacrimal

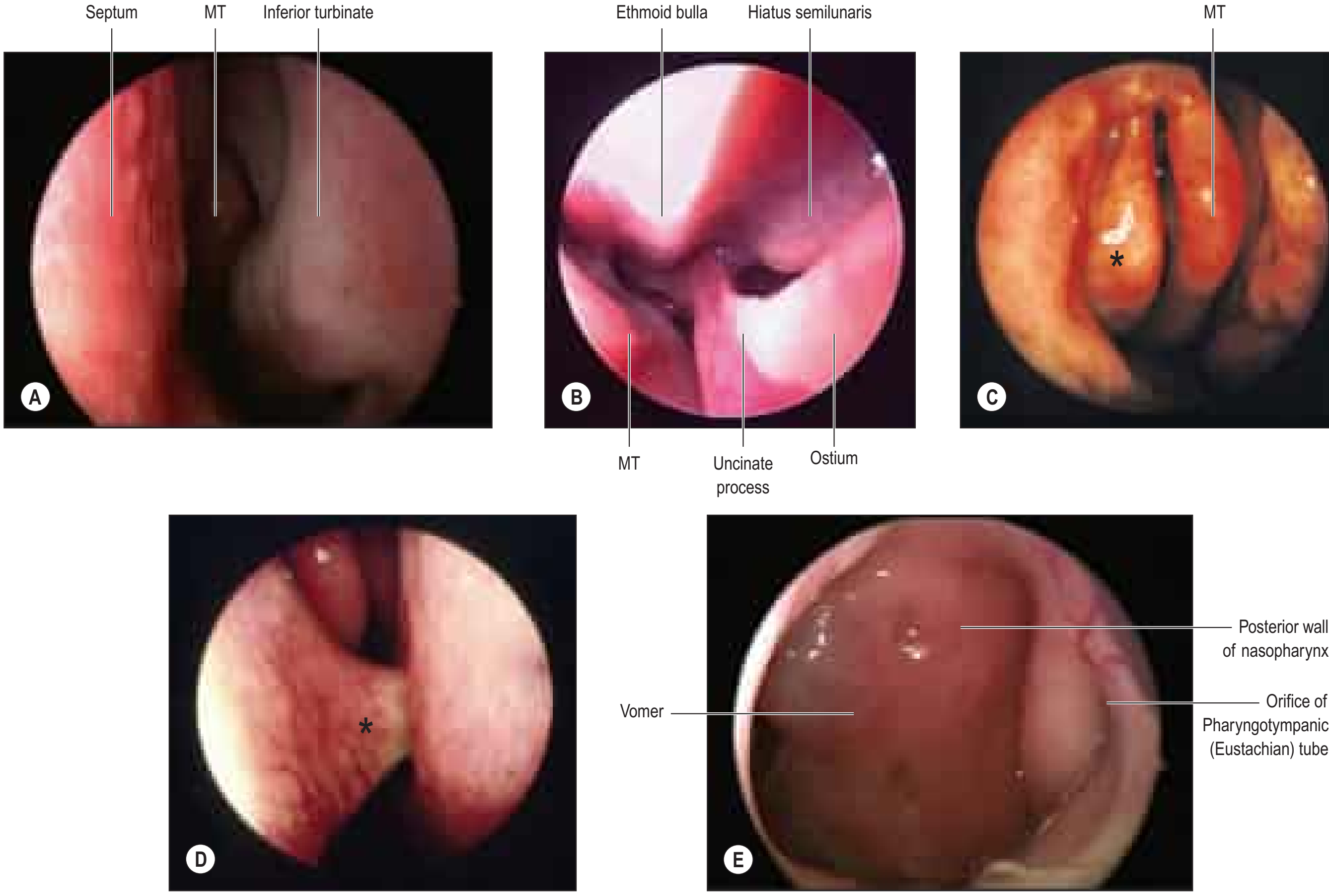

Fig. 33.7 **A**, An endoscopic view of the nasal cavity showing inferior and middle turbinates (MT). **B**, A view of the middle meatus and left lateral wall of the nose. **C**, An endoscopic view of the nose showing the middle turbinate and the pneumatized uncinate process in the middle meatus (asterisk). **D**, An endoscopic view of the nose demonstrating a deviated nasal septum (asterisk) making contact with the left inferior turbinate. **E**, An endoscopic view of the posterior nasal space.

canal. The canal is formed by the articulations between the lacrimal groove of the maxilla, the descending process of the lacrimal bone and the lacrimal process of the inferior concha. During postnatal development, the ostium of the nasolacrimal duct moves upwards and is increasingly hidden under the over-arching inferior concha. Inconsistent epithelial folds (the valve of Hasner) may remain at its distal opening.

Middle concha and middle meatus

The middle concha is a medial process of the ethmoidal labyrinth and may be pneumatized (conchal sinus). It extends back to articulate with the perpendicular plate of the palatine bone. The region beneath it is the middle meatus, which is deeper in front than behind, lies below and lateral to the middle concha, and continues anteriorly into a shallow fossa above the vestibule, termed the atrium of the middle meatus.

Sphenopalatine foramen

The sphenopalatine foramen, which is really a fissure, transmits the sphenopalatine artery and nasopalatine and superior nasal nerves from the pterygopalatine fossa. It is posterior to the middle meatus, and bounded above by the body and concha of the sphenoid, below by the superior border of the perpendicular plate of the palatine bone, and in front and behind by the orbital and sphenoidal processes of the palatine bone, respectively. The crista ethmoidalis, a small bony crest formed by the attachment of the basal lamella of the middle turbinate to the ascending palatine bone, is found anterior to the foramen, and is a reliable surgical landmark.

Ethmoturbinals

The ethmoturbinals are the superior and middle turbinates, occasionally supplemented by a supreme turbinate. They appear during weeks nine and ten of gestation as multiple folds on the developing lateral nasal wall and subsequently fuse into three or four ridges, each with an anterior (ascending) and a posterior (descending) ramus, separated by grooves. The first ridge develops into the agger nasi and the uncinate process. The second is thought to become the ethmoidal bulla and the fourth, if present, develops into the superior and supreme turbinates. The third is known as the basal lamella of the middle turbinate.

Attachments of the basal lamella of the middle turbinate

An understanding of the attachment of the middle turbinate to the roof and lateral wall of the nose is essential when undertaking sinus surgery. Anteriorly, it attaches to the crista ethmoidalis of the maxilla, and posteriorly it attaches to the crista ethmoidalis of the palatine bone, anterior to the sphenopalatine foramen. In between these points, the insertion lies in three different planes. The anterior third inserts vertically into the cranial base. The middle third turns laterally across the cranial base to the lamina papyracea, where it turns inferiorly; it may be indented by either anterior or posterior ethmoidal cells, but separates the two groups of cells. The posterior third runs horizontally, attaching to the lamina and the medial wall of the middle turbinate.

Superior concha and superior meatus

The superior concha is a medial process of the ethmoidal labyrinth and presents as a small curved lamina, posterosuperior to the middle concha. It roofs the superior meatus and is the shortest and shallowest of the three conchae. Above the superior concha, the sphenoidal sinus opens into the triangular sphenoethmoidal recess, which separates the superior concha and anterior aspect of the body of the sphenoid. The superior meatus is a short oblique passage extending about halfway along the upper border of the middle concha. The posterior ethmoidal sinuses open, via a variable number of apertures, into its anterior part.

Highest (supreme) nasal concha

Occasionally, a fourth concha, the highest or supreme nasal concha, appears on the lateral wall of the sphenoethmoidal recess. The passage immediately below it is called the supreme nasal meatus; it may contain an opening for the posterior ethmoidal sinus.

Functions of the nasal turbinates

In vivo, the conchae are covered with a thick, vascular, glandular soft tissue layer with pseudostratified columnar epithelium. They contain erectile tissue, linked to trigeminal innervation detecting airflow and temperature; congestion and decongestion of the venous sinusoids regulate nasal resistance. Turbinates are also essential for filtration, heating and humidification of inspired air. They direct airflow to the olfactory cleft, and some areas receive direct innervation from the olfactory bulb. Hypertrophy of the turbinates in allergy or environmental irritation results in nasal obstruction.

NASAL AIRFLOW AND THE NASAL CYCLE

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Nasal obstruction

Disturbances of the normal airflow pattern, whether produced by mucocutaneous or skeletal changes within the nose, affect normal nasal breathing and are usually perceived as some form of nasal obstruction.

The septum may be displaced by injury or by disproportionate growth of the cartilage that may cause it to bend; sometimes, the deviation may cause unilateral nasal obstruction. Variations in the anatomy of the lateral nasal wall, usually associated with variations in the size and position of the anterior ethmoidal cells, may obstruct frontal or maxillary sinus drainage, e.g. frontal cells, supraorbital ethmoidal cells, and infraorbital or Haller's cells, which represent an extension of anterior ethmoidal pneumatization along the infraorbital margin, sometimes within the roof of the maxillary sinus (see below).

NASAL AND OLFACTORY MUCOSAE

Nasal mucosa

The lining of the anterior part of the nasal cavity and vestibule is continuous with the skin, and consists of keratinized stratified squamous epithelium overlying a connective tissue lamina propria. Inferiorly, the skin bears coarse hairs (vibrissae), which curve towards the naris and help to arrest the passage of particles in inspired air. In males, after middle age, these hairs increase considerably in size. Further posteriorly, at the limen nasi, this changes into a mucosa, lined initially by non-keratinizing stratified squamous epithelium, and then by pseudostratified ciliated (respiratory) epithelium rich in goblet cells (see Fig. 2.2D). Respiratory epithelium forms most of the surface of the nasal cavity, i.e. it covers the conchae, meatuses, septum, floor and roof, except superiorly in the olfactory cleft, where the olfactory epithelium is present. It is adherent to the periosteum or perichondrium of the neighbouring skeletal structures. In some areas, cells of the respiratory epithelium may be low columnar or cuboidal, and the proportion of ciliated to non-ciliated cells is variable.

There are numerous seromucous glands within the lamina propria of the nasal mucosa. Their secretions make the surface sticky so that it traps particles in the inspired air. The mucous film is continually moved by ciliary action (the mucociliary escalator or rejection current) posteriorly into the nasopharynx at a rate of 6 mm per minute. Palatal movements transfer the mucus and its entrapped particles to the oropharynx for swallowing, but some also enters the nasal vestibule anteriorly. The secretions of the nasal mucosa contain the bacteriocides lysozyme, β-defensin and lactoferrin, and also secretory immunoglobulins (IgA). The mucosa is continuous with the nasopharyngeal mucosa through the posterior nasal apertures, the conjunctiva through the nasolacrimal duct and lacrimal canaliculi, and the mucosa of the sphenoidal, ethmoidal, frontal and maxillary sinuses through their openings into the meatuses.

The mucosa is thickest and most vascular over the conchae, especially at their extremities, and also on the anterior and posterior parts of the nasal septum and between the conchae. The mucosa is very thin in the meatuses, on the nasal floor and in the paranasal sinuses. Its thickness reduces the volume of the nasal cavity and its apertures significantly. The lamina propria contains cavernous vascular tissue with large venous sinusoids.

Olfactory mucosa

Olfactory mucosa (Fig. 33.8) covers approximately 5 cm^2 of the posterior upper parts of the lateral nasal wall, including the upper part of the vertical portion of the middle concha (where it is interspersed with

The paired nasal valves are critical regulators of nasal airflow and resistance. Each is divided into external and internal portions, although often only the internal portion is regarded as the 'nasal valve' or flow-limiting segment. The external valve is made up of the ala, the skin of the vestibule, the nasal sill and the contour of the medial crus of the lower lateral cartilage. The external valve has a tendency to collapse at high flow rates, even in normal individuals.

The upper lateral cartilages are in continuity with the superior border of the nasal septum. A valve-like mechanism exists at its distal end that regulates nasal airflow. The nasal valve area and internal nasal valve are two entities that should not be confused. The nasal valve area is the narrowest portion of the nasal passage. It is bounded medially by the septum and the tuberculum of Zuckerkandl, and superiorly and laterally by the caudal margin of the upper lateral cartilage, its fibroadipose attachment to the piriform aperture and the anterior end of the inferior turbinate; inferiorly, it consists of the floor of the piriform aperture. The nasal valve, on the other hand, is the specific slit-like segment between the caudal margin of the upper lateral cartilage and the septum. It is measured in degrees and normally ranges between 10 and 15°. Dilator naris anterior and the alar part of nasalis support the nasal valves.

Airflow dynamics are recognized as playing a key role in conditioning (i.e. warming, humidifying and filtering) inspired air; however, these dynamics are still poorly understood (Churchill et al 2004, Boyce and Eccles 2006). It is generally agreed that airflow within the nose is both laminar and turbulent. Inspired air is forced to pass through the nasal valve and then expands as it passes further into the nasal cavity, which offers little airflow resistance. This sudden change in speed and pressure produces turbulence and eddies. These currents allow adequate contact of inspired air with respiratory epithelium and facilitate odorant transport to the olfactory area. Chewing also affects nasal airflow; pulses of aroma-laden air are pumped out of the mouth into the retronasal region with each chew.

The cross-sectional area of the nasal airway depends on the dimensions of the septal partition and inferior turbinates (which are modulated by changes in the dimensions of the erectile tissue that covers them), and on the stability of the lateral nasal walls during breathing. The nasal cycle is an alternating fluctuation of nasal engorgement and airflow through the nasal passages, with period lengths ranging from 1 to 5 hours. The mechanism involves changes in sympathetic tone to the venous erectile tissue of the nasal mucosa; increased sympathetic vasoconstriction causes resistance to fall. These alterations in the thickness and contours of the mucosal surfaces are visible as a swelling or shrinkage of the nasal lining, and may serve to protect the mucosae from desiccation. The pacemaker for the cycle is believed to lie within the suprachiasmatic nucleus. The rhythmicity of the cycle decreases with age.

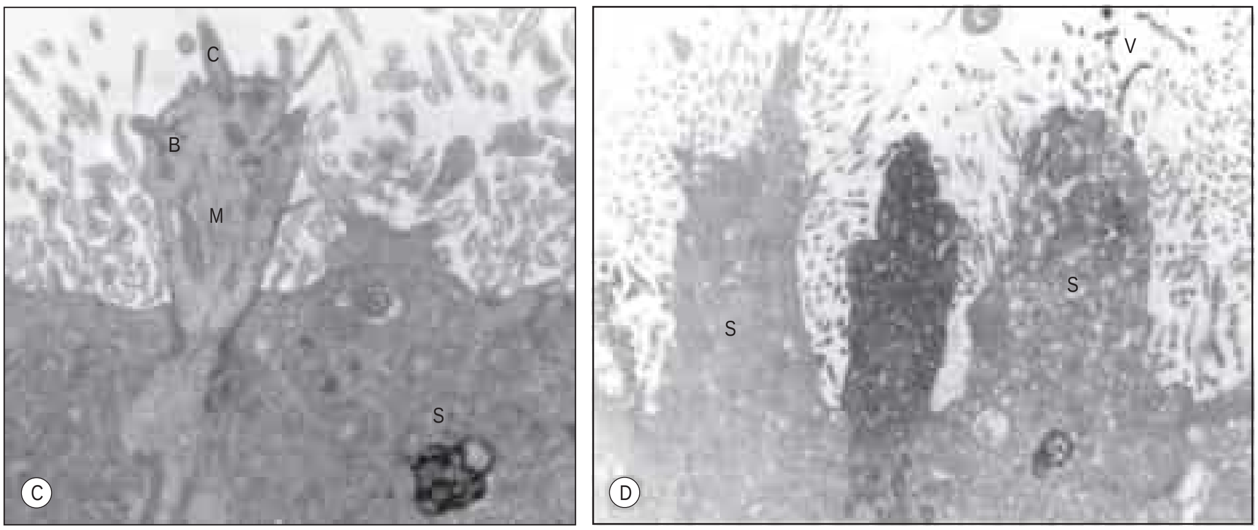

Fig. 33.8 **C**, A higher-power view of the expanded end of an olfactory receptor neurone. The electron-dense, osmiophilic material within the adjacent supporting cell (S) is thought to contribute to the pigmentation of the olfactory epithelium. Other abbreviations: B, basal body with projecting 'feet'; C, olfactory cilium; M, microtubules. **D**, A section of human olfactory epithelium immunostained with anti-OMP (olfactory marker protein). An immunopositive olfactory receptor neurone lies between two unstained supporting cells (S). Other abbreviations: V, microvilli.

CHAPTER 33

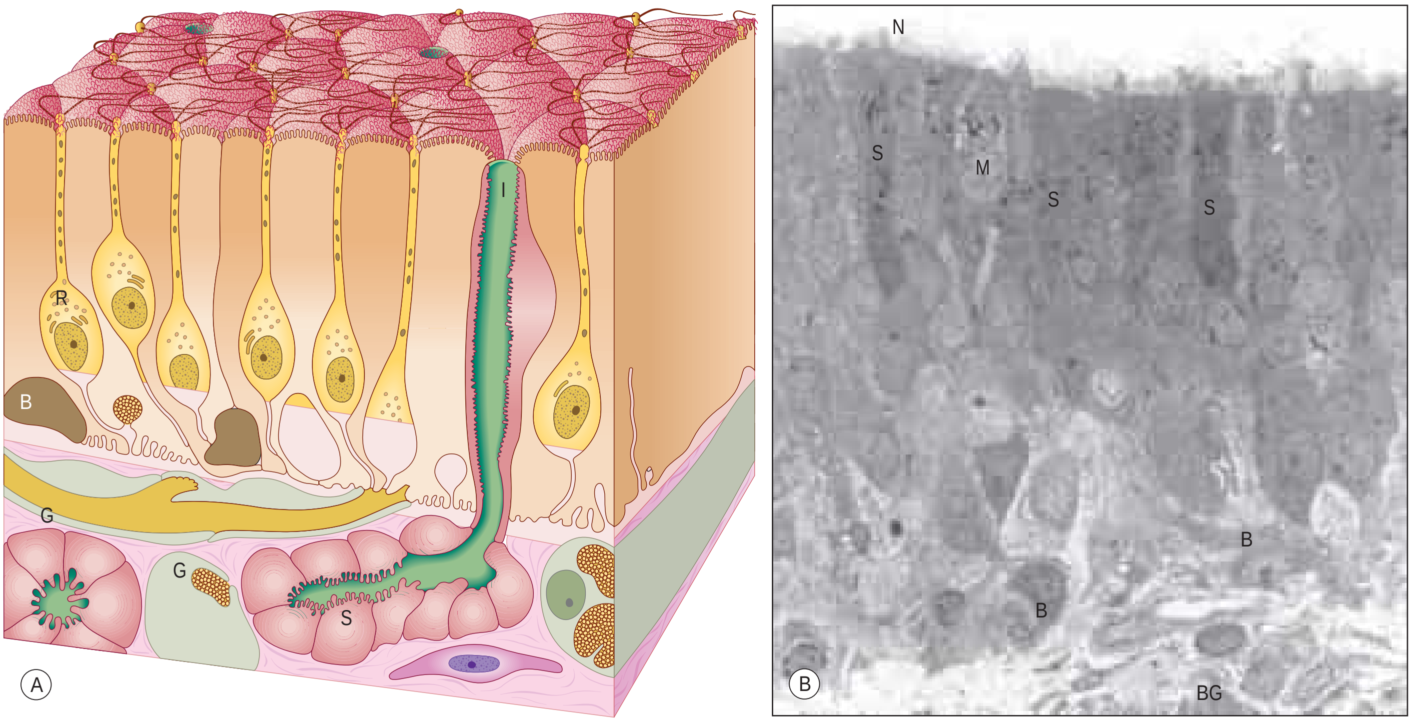

Fig. 33.8 **A**, The chief cytological features of the olfactory epithelium. Receptor cells (neurones) (R) are situated among columnar sustentacular cells. The axons of the receptor cells emerge from the epithelium in bundles enclosed by ensheathing glial cells (G). Rounded globose basal cells (B) and flattened horizontal basal cells (not shown) lie on the basal lamina, and the subepithelial glands (of Bowman) (S) open on to the surface via their intraepithelial ducts (I). At the surface are cilia of the receptor cells and microvilli of the supporting cells. **B–D**, Longitudinal sections through human olfactory epithelium. **B**, Ciliated olfactory receptor neurones with characteristic expanded ends (N) project into the nasal lumen. The edge of a Bowman's gland (BG) lies deeper in the lamina propria. Other abbreviations: B, basal cells resting on the basal lamina; M, microvillar cell; S, supporting or sustentacular cells containing electron-dense material. (B–D, Courtesy of Professor Bruce Jafek, Department of Otolaryngology, University of Colorado, Denver, USA.)

respiratory epithelium in a chequerboard fashion) and the opposite part of the nasal septum, the superior concha, the sphenoethmoidal recess, the upper part of the perpendicular plate of the ethmoid and the portion of the roof of the nose that arches between the septum and lateral wall, including the underside of the cribriform plate (constituting the olfactory cleft or groove). It consists of a yellowish-brown pigmented pseudostratified epithelium, containing olfactory receptor neurones, sustentacular cells and two classes of basal cell, lying on a subepithelial lamina propria containing subepithelial olfactory glands (of Bowman) and bundles of axons derived from the olfactory receptor neurones that course through the mucosa on their way to the cribriform plate. The glands secrete a predominantly serous fluid through ducts that open on to the epithelial surface. These secretions form a thin fluid layer in which sensory cilia and the microvilli of the sustentacular cells are embedded.

Olfactory receptor neurones Olfactory receptor neurones are bipolar. Their cell bodies and nuclei are located in the middle zone of the olfactory epithelium. Each neurone has a single unbranched apical dendrite, 2 μm in diameter, which extends to the epithelial surface; and a basally directed unmyelinated axon, 0.2 μm in diameter, which passes in the opposite direction, penetrates the basal lamina and enters the lamina propria. The tips of the dendrites project into the overlying secretory fluid and are expanded into characteristic endings (knobs) (see Fig. 33.8B). Groups of up to 20 cilia radiate from the circumference of each ending and extend for long distances parallel to the epithelial surface. Internally, the short proximal part of each cilium has the '9+2' pattern of microtubules typical of motile cilia, while the longer distal trailing end contains only the central pair of microtubules. The olfactory cilia lack dynein arms and are thought to be non-motile; their primary purpose is to increase the surface area of sensory receptor membrane available for the efficient detection of odorant molecules transferred across the mucous layer by odorant-binding proteins. Mature olfactory neurones express olfactory marker protein (OMP), an abundant cytoplasmic protein involved in olfactory signal transduction (see Fig. 33.8D). Each olfactory receptor neurone expresses receptors for a single odorant molecule (or very few). In humans, over 1000 genes code for functional odorant receptors; the number of functional genes is much higher in macrosmotic animals (Buck and Axel 1991). Although neurones with the same receptor specificity are randomly distributed within anatomical zones of the epithelium, their axons all converge on the same glomerulus in the olfactory bulb. Specific odours activate a unique spectrum of receptor neurones, which in turn activate restricted groups of glomeruli and their second-order neurones.

The axons form small intraepithelial fascicles among the processes of sustentacular and basal cells. The fascicles penetrate the basal lamina and are immediately surrounded by olfactory ensheathing cells. Groups of up to 50 such fascicles join to form larger olfactory nerve rootlets that pass through the cribriform plate of the ethmoid bone, wrapped in meningeal sheaths. They immediately enter the overlying olfactory bulbs, where they synapse in glomeruli with mitral cells and, to a lesser extent, with smaller tufted cells.

Microvillar cells Microvillar cells occupy a superficial position in the olfactory epithelium. They are flask-shaped and electron-lucent, and the apical end of each cell gives rise to a tuft of microvilli that project into the mucus layer lining the nasal cavity (see Fig. 33.8B). Cell counts in longitudinal sections reveal that microvillar cells occur with a density that is approximately one-tenth of the density of ciliated olfactory neurones; their function and origin have yet to be determined.

Sustentacular cells Sustentacular, or supporting, cells are columnar cells that separate and partially ensheathe the olfactory receptor neurones. Their large nuclei form a layer superficial to the neuronal nuclei within the epithelium. The cells are capped by numerous long, irregular, microvilli, which lie in the secretory fluid layer that covers the surface of the epithelium, intermingled with the trailing ends of the cilia on the olfactory receptor endings. Their expanded bases contain numerous lamellated dense bodies, which are the remnants of secondary lysosomes, and which contribute significantly to the pigmentation of the olfactory area (see Fig. 33.8B,C). The granules gradually accumulate with age, and because these cells are long-lived, the intensity of pigmentation also increases with age. Neighbouring sustentacular cells are linked by desmosomes close to the epithelial surface, an arrangement that helps to stabilize the epithelium mechanically. Sustentacular cells and olfactory receptor neurones are linked by tight junctions at the level of the epithelial surface.

Basal cells There are horizontal and globose basal cells. Horizontal basal cells are flattened against the basal lamina. Their nuclei are condensed and their darkly staining cytoplasm contains numerous intermediate filaments of the cytokeratin family, inserted into desmosomes between the basal cells and surrounding sustentacular cells. Globose cells are rounded or elliptical in shape, and have pale, euchromatic nuclei and pale cytoplasm. They form a distinct zone that is slightly internal to the basal surface of the epithelium and characterized by mitotic figures; globose basal cells are the immediate source of new olfactory receptor neurones.

Olfactory ensheathing cells Olfactory ensheathing cells share properties with astrocytes and non-myelinating Schwann cells, but also possess distinctive features that indicate they are a separate class of glia. Developmentally, they are derived from the olfactory placode rather than the neural crest. They ensheathe olfactory axons in a unique manner throughout their entire course and accompany them into the olfactory bulb, where they contribute to the glia limitans. In recent years, olfactory ensheathing cells have been the focus of intense experimental scrutiny in the search for a source of transplantable glia capable of supporting neuronal regeneration within the central nervous system, possibly in the treatment of paraplegia.

Olfactory glands Olfactory (Bowman's) glands are branched tubulo-alveolar structures that lie beneath the olfactory epithelium and secrete their products on to the epithelial surface through narrow, vertical ducts. Their secretions, which include defensive substances, lysozyme, lactoferrin, IgA and sulphated proteoglycans, together with odorant-binding proteins which increase the efficiency of odour detection, bathe the dendritic endings and cilia of the olfactory receptors. The fluid acts as a solvent for odorant molecules, allowing their diffusion to the sensory receptors.

Turnover of olfactory receptor neurones Olfactory receptor neurones are lost and replaced throughout life. Individual receptor cells have a variable lifespan, thought to average 1–3 months. Stem cells situated near the base of the epithelium undergo periodic mitotic division throughout life, giving rise to new olfactory receptor neurones, which then grow a dendrite to the olfactory surface and an axon to the olfactory bulb. The cell bodies of these new receptor neurones gradually move apically until they reach the region just below the supporting cell nuclei. When they degenerate, dead neurones either are shed from the epithelium or are phagocytosed by sustentacular cells. The rate of receptor cell loss and replacement increases after exposure to damaging stimuli but declines slowly with age, a phenomenon that presumably contributes to diminishing olfactory sensory function in old age. Biopsy specimens from normosmic adults have revealed that patchy replacement of olfactory with respiratory epithelium occurs even in young healthy adults (Paik et al 1992, Holbrook et al 2005).

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE OF THE NASAL CAVITY

Many of the vessels and nerves supplying the nasal cavities arise within the pterygopalatine fossa and these origins are described in Chapter 32.

Arteries

Branches of the ophthalmic, maxillary and facial arteries supply different territories within the walls, floor and roof of the nasal cavity (Fig. 33.9). They ramify to form anastomotic plexuses within and deep to the nasal mucosa. Anastomoses also occur between some larger arterial branches. The anterior and posterior ethmoidal branches of the ophthalmic artery supply the ethmoidal and frontal sinuses and the roof of the nose (including the septum). The anterior ethmoidal artery usually runs within the bone of the anterior skull base, unless this is well pneumatized with a supraorbital cell, in which case the artery is more likely to be positioned away from the skull base and to be more prone to surgical damage. The sphenopalatine branch of the maxillary artery supplies the mucosa of the turbinates, meatuses and posteroinferior part of the nasal septum, i.e. it is the principal vessel supplying the nasal mucosa. The artery comes out of a fissure (erroneously termed a foramen), and normally divides before it enters the nasal cavity behind the crista ethmoidalis into posterior lateral nasal and posterior septal branches. Sometimes, the artery may divide before it leaves the foramen. The number and distribution of its branches show great variation, with a median of three or four branches (Babin et al 2003). The greater palatine branch of the maxillary artery supplies the region of the inferior meatus. A branch crosses the sphenoidal rostrum, below its natural ostium, to supply the nasal septum. This is utilized to provide pedicled, vascularized nasoseptal flaps in skull-base reconstruction. Its terminal part ascends through the incisive canal to anastomose on the septum with branches of the sphenopalatine and anterior ethmoidal arteries, and with the septal branch of the superior labial artery. This septal region (Little's area or Kiesselbach's plexus) is a common site of bleeding from the nose. The infraorbital artery and the superior, anterior and posterior alveolar branches of the maxillary artery supply the mucosa of the maxillary sinus. The pharyngeal branch of the maxillary artery supplies the sphenoidal sinus.

Nose bleeds

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CHAPTER 33

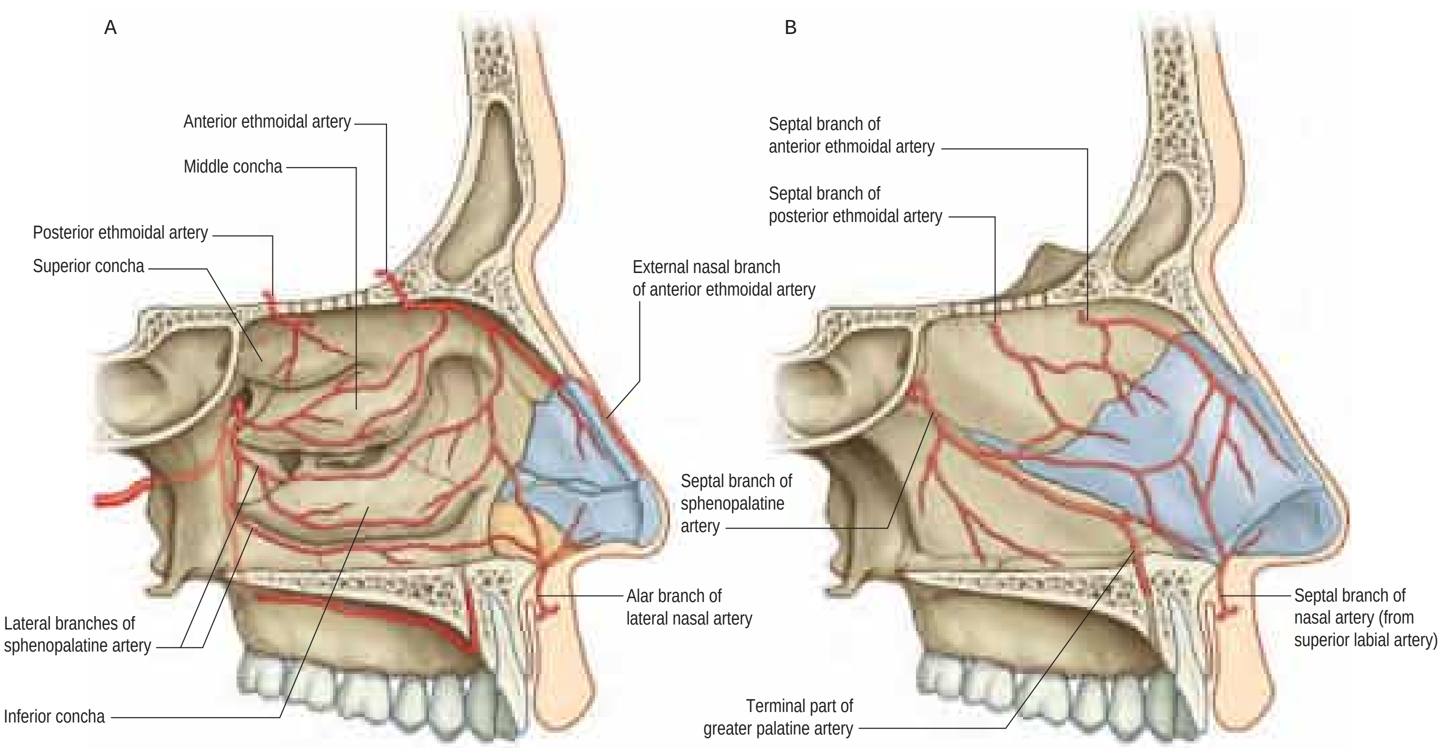

Fig. 33.9 The arterial supply of the nasal cavity. **A**, The lateral wall of the left nasal cavity. **B**, The medial wall of the left nasal cavity. (From Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

The vast majority of nose bleeds, particularly in children, occur as a result of digital trauma to the anastomosis of arterioles and veins in Little's area, on the nasal septum just inside the nasal vestibule. This area is amenable to cautery when required. In older patients, brisker bleeding may occur as a result of the spontaneous rupture of arteries further back in the nose. These may be controlled by applying pressure with a nasal pack, but where this fails, knowledge of the pattern of arterial blood supply to the nasal cavity permits interruption of the appropriate blood supply by ligation or embolization of the feeding vessel. The sphenopalatine artery may be ligated under endoscopic visualization as it enters the nose through the sphenopalatine foramen. The ethmoidal arteries may be exposed within the orbit and ligated to arrest bleeding high up in the nasal cavity. The maxillary artery may be exposed surgically behind the posterior wall of the maxillary sinus and ligated, or alternatively it may be identified radiologically, using a radiopaque dye, so that it may be blocked by embolization (Simmen and Jones 2010).

Veins

A rich submucosal cavernous plexus is especially dense in the posterior part of the septum and in the middle and inferior turbinates (Fig. 33.10). Numerous arteriovenous anastomoses are present in the deep layer of the mucosa and around the mucosal glands. The cavernous turbinate plexuses resemble those in erectile tissue; the nasal cavity is susceptible to blockage, should they become engorged. Veins from the posterior part of the nose generally pass to the sphenopalatine vein that runs back through the sphenopalatine foramen to drain into the pterygoid venous plexus. The anterior part of the nose is drained mainly through veins accompanying the anterior ethmoidal arteries, and these veins subsequently pass into the ophthalmic or facial veins. Injection of vasoconstrictive agents or corticosteroids during surgery, particularly to the inferior turbinates, may permit access to the intracranial and ophthalmic circulations. Blindness has been reported in rare cases following such injections. A few veins pass through the cribriform plate to connect with those on the orbital surface of the frontal lobes of the brain. When the foramen caecum is patent, it transmits a vein from the nasal cavity to the superior sagittal sinus.

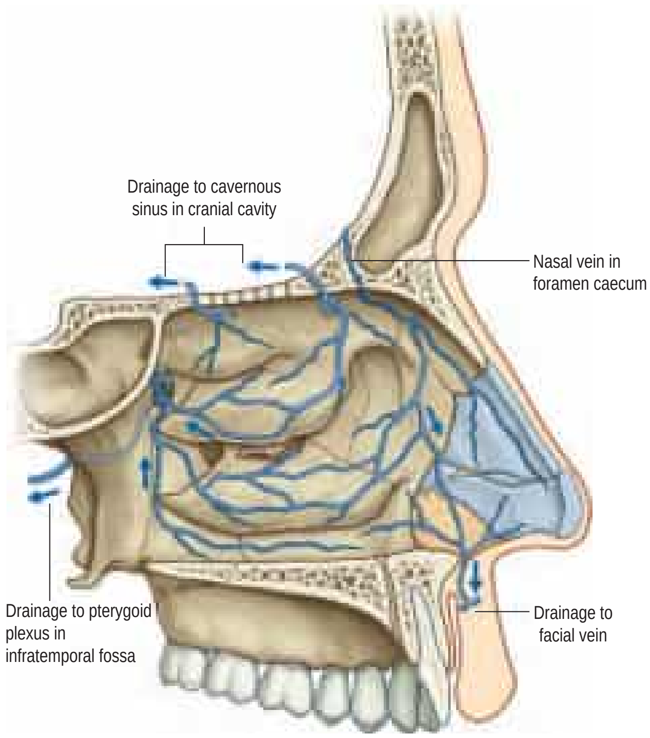

Fig. 33.10 The venous drainage of the nasal cavity: the lateral wall of the left nasal cavity. (From Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

Lymphatic drainage

Lymph vessels from the anterior region of the nasal cavity pass superficially to join those draining the external nasal skin and end in the submandibular nodes. The rest of the nasal cavity, paranasal sinuses, nasopharynx and pharyngeal end of the pharyngotympanic tube all drain to the upper deep cervical nodes, either directly or through the retropharyngeal nodes. The posterior nasal floor probably drains to the parotid nodes.

INNERVATION OF THE NASAL CAVITY

Olfaction is mediated via the olfactory nerves. General sensation (touch, pain and temperature) from the nasal mucosa is carried by branches of the ophthalmic and maxillary divisions of the trigeminal nerves (Fig. 33.11). Trigeminal fibres close to, and within, the epithelial layer are sensitive to noxious chemicals, e.g. ammonia and sulphur dioxide. Autonomic fibres innervate mucous glands and control cyclical and reactive vasomotor activity.

Trigeminal innervation

The anterior ethmoidal branch of the nasociliary nerve leaves the cranial cavity through a small slit near the crista galli and enters the roof of the nasal cavity, where it runs in a groove on the inner surface of the nasal bone, supplying the roof of the nasal cavity. It gives off a lateral internal branch to supply the anterior part of the lateral wall, and a medial internal branch to the anterior and upper parts of the septum, before emerging at the inferior margin of the nasal bone as the external nasal nerve to supply the skin of the external nose to the nasal tip; damage following nasal trauma may result in paraesthesia of the tip. The infraorbital nerve supplies the nasal vestibule. The anterior superior alveolar nerve supplies part of the septum, the floor near the anterior

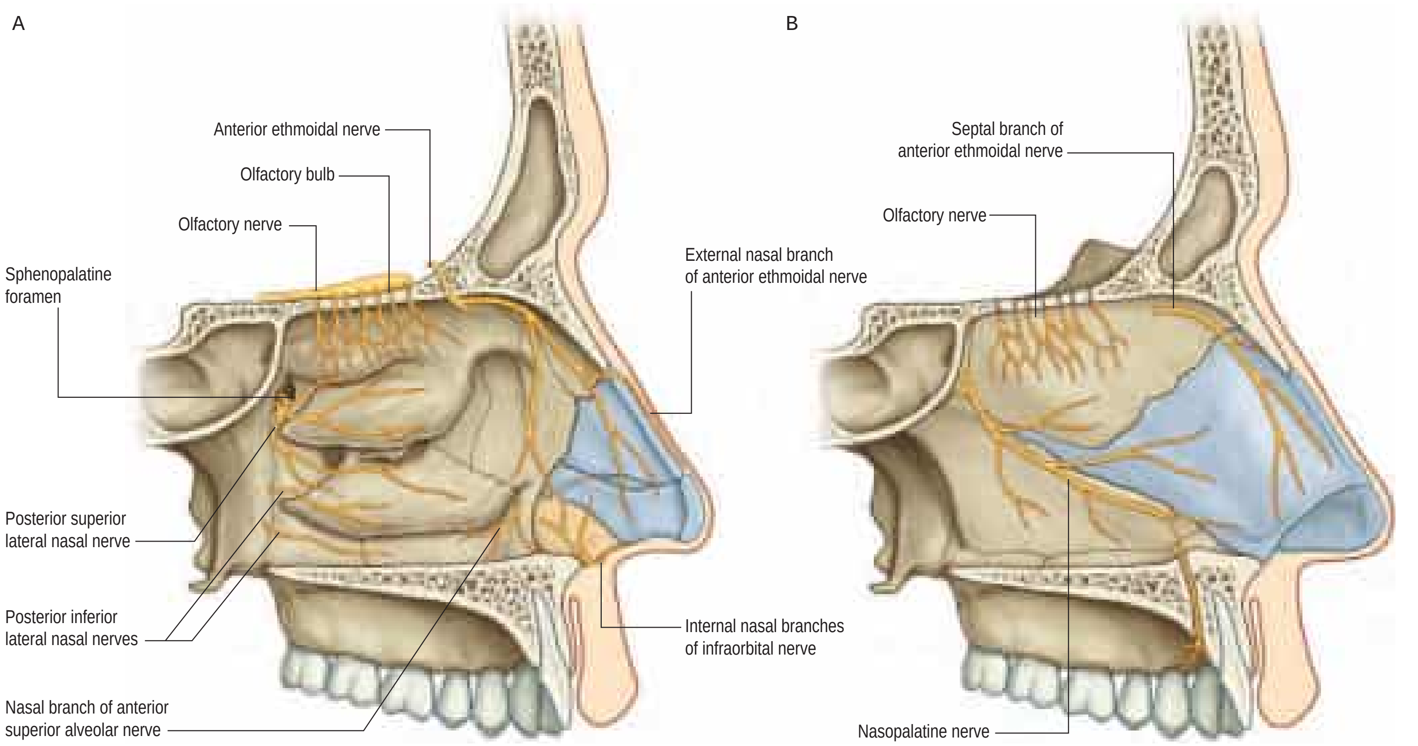

Fig. 33.11 The innervation of the nasal cavity. **A**, The lateral wall of the left nasal cavity. **B**, The medial wall of the left nasal cavity. (From Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

nasal spine and the anterior part of the lateral wall as high as the opening of the maxillary sinus; the lateral posterior superior nasal and the posterior inferior nasal branches of the greater palatine nerve together supply the posterior three-quarters of the lateral wall, roof and floor; the medial posterior superior nasal nerves and the nasopalatine nerve supply the inferior part of the nasal septum; and branches from the nerve of the pterygoid canal supply the upper and posterior part of the roof and septum.

Autonomic innervation

The deep petrosal nerve (postganglionic sympathetic fibres) and the greater petrosal nerve (preganglionic parasympathetic fibres) converge to form the nerve of the pterygoid canal (Vidian nerve in the Vidian canal). The canal enters the pterygopalatine fossa, and the nerve joins the pterygopalatine ganglion, where the parasympathetic fibres synapse, but the sympathetic fibres pass through without synapsing (Ch. 32). The nerve of the pterygoid canal is an important landmark to the petrous portion of the internal carotid artery, and may also rarely be transected to treat intractable rhinorrhoea. Sympathetic postganglionic vasomotor fibres are distributed to the nasal blood vessels. Postganglionic parasympathetic fibres derived from the pterygopalatine ganglion provide the secretomotor supply to the nasal mucous glands, and are distributed via branches of the maxillary nerves.

Olfactory nerves

Olfactory nerves are bundles of very small axons derived from olfactory receptor neurones in the olfactory mucosa. The axons are unmyelinated, and in varying stages of maturity, reflecting the constant turnover of olfactory neurones that takes place in the olfactory epithelium. Bundles of axons surrounded by olfactory ensheathing cells form a plexiform network in the subepithelial lamina propria of the mucosa. The bundles unite into as many as 20 branches that cross the cribriform plate in lateral and medial groups, and enter the overlying olfactory bulb, where they end in glomeruli. Each branch is ensheathed by dura mater and pia arachnoid as it passes through the cribriform plate (Fig. 33.12). The dura subsequently becomes continuous with the nasal periosteum, and the pia arachnoid merges with the connective tissue sheaths surrounding the nerve bundles, an arrangement that may favour the spread of infection into the cranial cavity from the nasal cavity.

In severe injuries involving the anterior cranial fossa, the olfactory bulb may be separated from the olfactory nerves or the nerves may be torn, producing anosmia, i.e. loss of olfaction. Fractures may involve the meninges, so that cerebrospinal fluid may leak into the nose, resulting in cerebrospinal rhinorrhoea. Such injuries open up avenues for intracranial infection from the nasal cavity.

Vomeronasal organ

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PARANASAL SINUSES

The paranasal sinuses are the frontal, ethmoidal, sphenoidal and maxillary sinuses, housed within the bones of the same name (see Fig. 33.1; Figs 33.13–33.15). They all open into the lateral wall of the nasal cavity by small apertures that permit both the equilibration of air between the various air spaces and the clearance of mucus from the sinuses into the nose via a mucociliary escalator. The detailed position of these apertures, and the precise form and size of each of the sinuses, vary enormously between individuals (Lang 1989, Beale et al 2009, Navarro 1997). Respiratory epithelium extends through the apertures of the paranasal sinuses to line their cavities, a feature that unfortunately favours the spread of infections. Sinus mucosa is thinner and less vascular, and has fewer goblet cells, than nasal mucosa. Cilia are always present in the mucosa near the apertures but less evenly distributed elsewhere within the sinuses.

The functions of the paranasal sinuses remain speculative. They clearly add some resonance to the voice, and also allow the enlargement of local areas of the skull while minimizing a corresponding increase in bony mass. It is likely that such growth-related changes serve to strengthen particular regions, e.g. the alveolar processes of the maxillae when the secondary dentition erupts, but they may also function in

Fig. 33.12 Axial (**A**, **B**) and sagittal (**C**) scans through the olfactory groove. **A**, Computed tomogram (CT). **B**–**C**, Magnetic resonance fast imaging employing steady-state acquisition (FIESTA) sequence. Key: 1, ethmoid plate; 2, inferior orbital fissure; 3, olfactory bulb. (With permission from Linn J Cranial nerves. In: Naidich TP, Castillo M, Cha S, Smirniotopoulos JG (eds) Imaging of the Brain. 2013, Elsevier, Saunders.)

CHAPTER 33

In most amphibians, reptiles and mammals, the vomeronasal organ is the peripheral sensory organ of the accessory olfactory system. In these animals, paired vomeronasal organs are located either at the base of the nasal septum or in the roof of the mouth, and are involved in chemical communication that is often, but not exclusively, mediated via pheromones. In many macrosomatic animals, the vomeronasal organ consists of a vomeronasal duct that contains chemosensory cells, and a vomeronasal nerve that terminates centrally in the accessory olfactory bulb. The vomeronasal organ exists in the developing human fetus but its existence in the adult has long been controversial. A bilateral structure in the anteroinferior wall of the nasal septum just dorsal to the paraseptal cartilages, which is presumed to be analogous to the vomeronasal organ of other vertebrates, has been demonstrated in adult human tissue; while there is no evidence that this structure has any neuroreceptive function, it has been suggested that it might have an as yet unknown endocrine function (Wessels et al 2014).

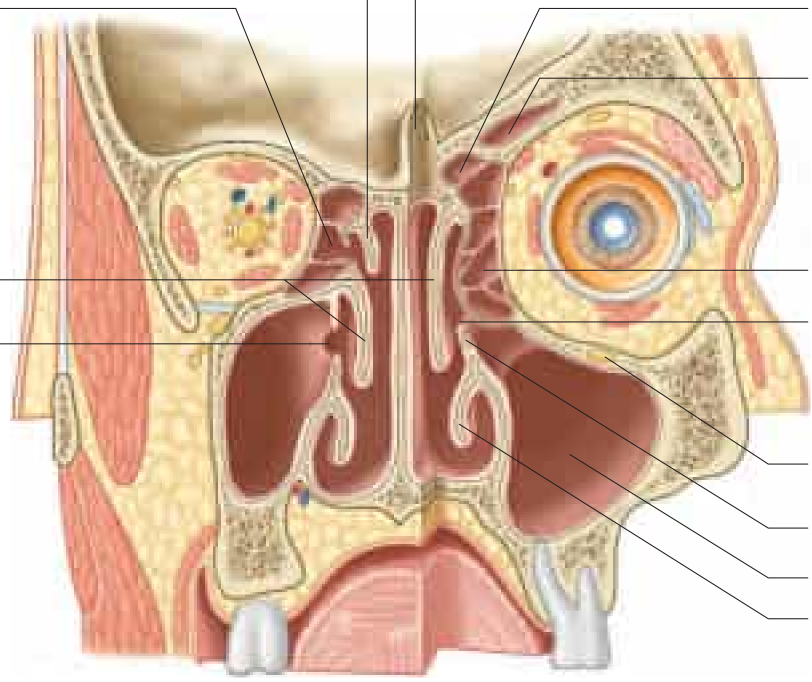

Fig. 33.13 A coronal section through the nasal cavity, viewed from the posterior aspect. The plane of the section on the right side is more anterior. The normal orifice of the maxillary sinus is shown on the right side and an accessory orifice on the left side.

Fig. 33.14 A coronal CT scan showing the sphenoidal air sinus. Abbreviations: C, posterior choanae; S, sphenoidal sinus.

contouring the head to provide visual signals indicating the individual's status in a social context (e.g. gender, sexual maturity and group identity).

Most sinuses are rudimentary or absent at birth, but enlarge appreciably during the eruption of the permanent teeth and after puberty, events that significantly alter the size and shape of the face.

DEVELOPMENT OF THE PARANASAL SINUSES, AND ANATOMICAL VARIATIONS IN CHILDHOOD

At birth, both small ethmoidal and maxillary sinuses are present, but the frontal sinus is nothing more than an out-pouching from the nasal cavity, and there is no pneumatization of the sphenoid bone. Understanding the development of the sinuses at each stage of childhood is essential for interpreting pathology and planning surgery. Cadaveric and radiological studies have provided normative data for sinus development.

Birth

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1–4 years

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4–8 years

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8–12 years

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Variations in sinus development

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FRONTAL SINUS

The paired frontal sinuses are posterior to the superciliary arches, between the outer and inner tables of the frontal bone (see Figs 33.13, 33.15A, B). Each usually underlies a triangular area on the surface of the face, its angles formed by the nasion, a point 3 cm above the nasion and the junction of the medial third and lateral two-thirds of the supraorbital margin. The two sinuses are rarely symmetric, since the septum between them usually deviates from the median plane. Each sinus may be further divided into a number of communicating recesses by incomplete bony septa.

The average dimensions of an adult frontal sinus are: height 3.2 cm; breadth 2.6 cm; and depth 1.8 cm. Each usually has a frontal portion that extends upwards above the medial part of the eyebrow, and an orbital portion that extends back into the medial part of the roof of the orbit. One or both sinuses may rarely be hypoplastic or even absent; racial differences have been reported. The prominence of the superciliary arches is no indication of the presence or size of the frontal sinuses.

The turbinates at birth are usually bulky and the inferior and middle meatuses are underdeveloped. The uncinate, hiatus semilunaris and ethmoidal bulla are already well-defined, fixed landmarks, and both the anterior and posterior ethmoidal cells are already almost completely developed in terms of number but not size. The cells are separated by connective tissue, which becomes compressed with subsequent expansion of the cells. The ethmoid complex ranges from 8 to 12 mm in length, 1 to 3 mm in width and 1 to 5 mm in height; the complexes expand rapidly in size in the first few years of life. The maxillary sinus is roughly spherical, with a volume of 6–8 cm^3, and measures 10 mm in length, 4 mm in height and 3 mm in width. It lies initially medial to the orbit, but projects laterally under the orbit by the end of the first year of life. The sphenoid is devoid of air, although a blind mucosal sac may sometimes be identified. At birth, the sphenoid has two major ossification centres; failure of fusion between the two may give rise to anatomical variations such as a persistent 'craniopharyngeal canal'. The Eustachian tube is found within the nasal cavity, behind the posterior end of the inferior turbinate. The frontal sinus is no more than a small out-pouching that drains into the infundibulum.

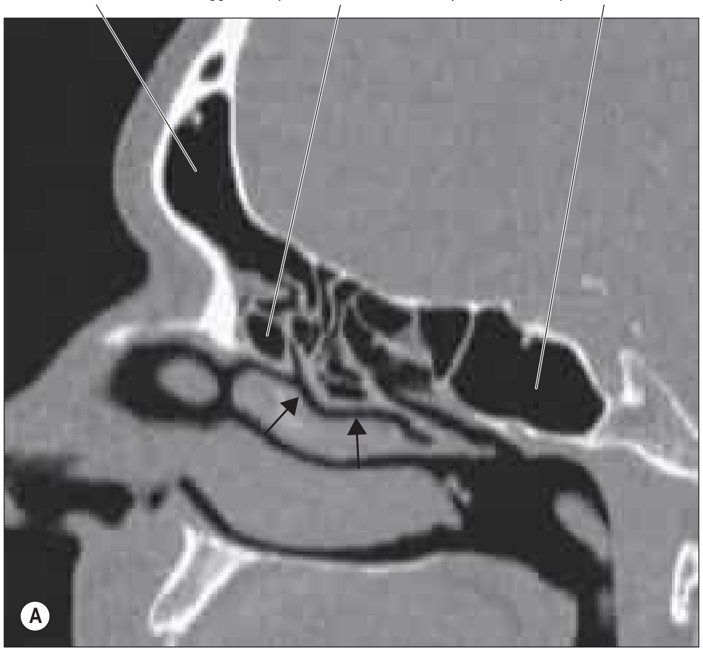
The supreme turbinate has usually disappeared, while the remaining three turbinates reduce relatively in size. The inferior meatus develops and contributes to the nasal airway. The maxillary sinus enlarges rapidly up to the age of 4 years, reaching laterally as far as the infraorbital canal, and inferiorly to the attachment of the inferior turbinate; it ranges between 22 and 30 mm in length, 12 and 18 mm in height and 11 and 19 mm in width. The ethmoidal cells enlarge in all directions, starting anteriorly and then progressing posteriorly. By the age of 4, they are well developed; variants such as the agger nasi cell, infraorbital cell and concha bullosa are identifiable. Sphenoidal pneumatization commences around 7 months of age; a distinct cell is visible by the age of 2 years. By 4 years of age, it is roughly pea-sized, with a diameter of 4–8 mm. The frontal sinus is the last to develop and is imperceptible in infants less than 1 year old. It begins to pneumatize after the age of 2, gradually enlarging as an out-pouching from the anterior ethmoids. Early growth is slow; by 4 years, the vertical height reaches only half the height of the orbit (between 6 and 9 mm in height) (Wolf et al 1993).

By 8 years, the sinuses have expanded in all directions. The maxillary sinus has reached the maxillary bone laterally and the plane of the hard palate, becoming tetrahedral in shape. It ranges from 34 to 38 mm in length, 22 to 26 mm in height and 18 to 24 mm in width. The dental buds of unerupted teeth are closely related to the floor of the sinus. The ethmoidal cells continue to enlarge, but more slowly than before; the posterior cells become larger than the anterior cells.

Pneumatization into the vertical plate of the frontal bone does not usually begin until the fifth year, at which time the frontal sinus may be identified radiologically. The frontal sinus pneumatizes rapidly and begins to pneumatize into the vertical plate of the frontal bone; its height reaches the orbital roof at 8 years (Ruf and Pancherz 1996).

By 12 years, the sinuses have almost reached adult proportions. The maxillary sinus pneumatizes into the maxillary alveolus after eruption of the permanent dentition, so that the floor of the sinus now sits 4–5 mm below the level of the floor of the nasal cavity. The ethmoidal sinuses reach adult size, and the frontal sinuses extend into the frontal bone, continuing to enlarge until puberty. The sphenoidal sinus reaches its full size by the age of 14 years.

Asymmetry in the size and shape of the sinuses, hypoplasia and anatomical variants are common (Navarro 1997). Unilateral or bilateral maxillary hypoplasia occurs infrequently; the condition is rarely bilateral. Pneumatization of an ethmoidal cell into the middle concha creates a concha bullosa, and inferolaterally creates an infraorbital cell. The degree of pneumatization of the sphenoid is highly variable but aplasia is very rare. In contrast, unilateral aplasia of the frontal sinus is present in 15% of adults, and present bilaterally in 5%.



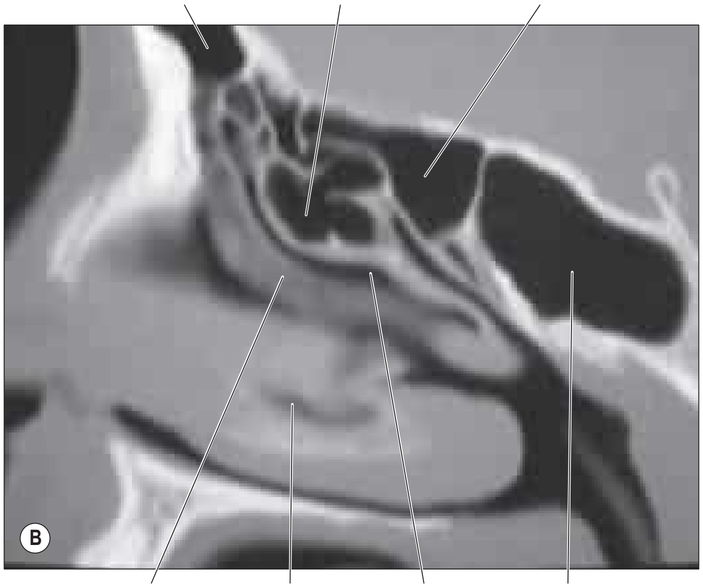


Fig. 33.15 **A**–**B**, Sagittal CT scans showing the frontal, ethmoidal and sphenoidal sinuses. Arrows indicate the hiatus semilunaris in **A**. (A, Courtesy of Professor D Bell.)

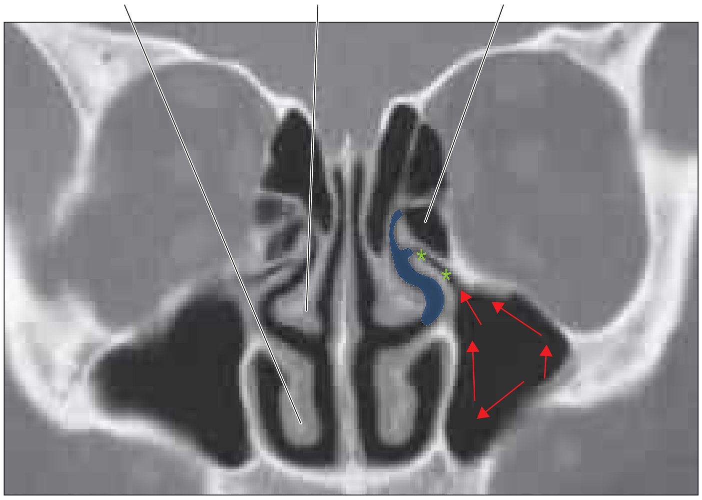


Fig. 33.17 A coronal CT scan through the ostiomeatal complex. Red arrows indicate the direction of mucociliary flow; the blue area, the middle meatus; and the green stars, the infundibulum.

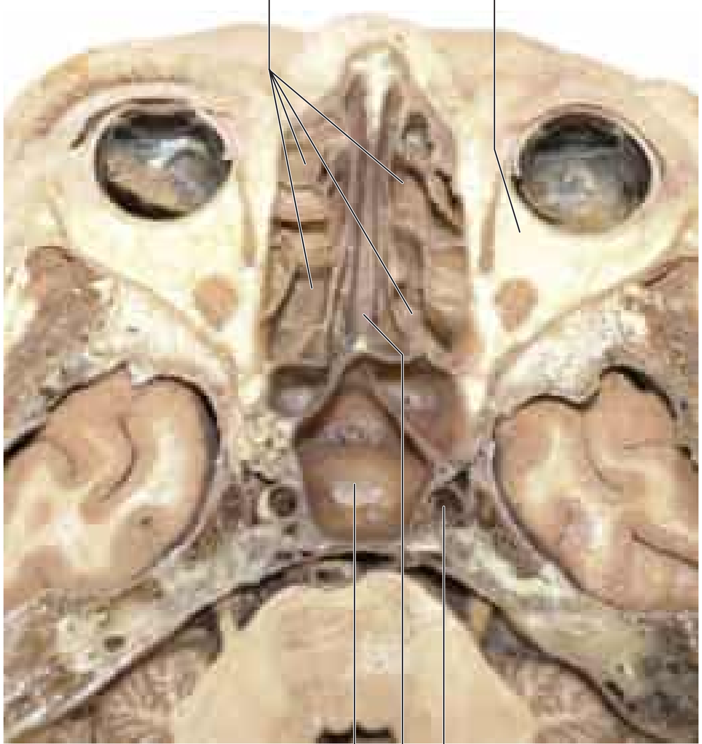


Fig. 33.18 A horizontal section of the head showing the ethmoidal and sphenoidal sinuses. (With permission from Berkovitz BKB, Moxham BJ 2002 Head and Neck Anatomy. London: Martin Dunitz.)

They are more prominent in males, and lend the forehead an obliquity that contrasts with the vertical or convex profile typical of children and females.

The aperture of each frontal sinus opens either into the anterior part of the corresponding middle meatus by the ethmoidal infundibulum as a frontonasal recess (rather than a duct), or medial to the hiatus semilunaris if the uncinate process is attached to the lateral nasal wall or an agger nasi cell (Kuhn 2002). The frontal recess is actually the most anterior part of the anterior ethmoidal complex but is described here because of its importance in the drainage of the frontal sinus. Its lateral wall is the lamina papyracea; the medial wall is formed by the middle turbinate; and posteriorly, the wall is made up of either the cranial base, in a suprabullar recess, or the insertion of the bulla, if this reaches the cranial base. Anteriorly, the wall extends from the frontal sinus proper to the anterior attachment of the middle turbinate. In its simplest form, it takes the shape of an inverted funnel, forming an hourglass shape with the floor of the frontal sinus. However, the frontal recess is often narrowed both from in front and from behind by anterior ethmoidal cells, such that it may be highly convoluted (Fig. 33.16). These fronto-ethmoidal cells are classified with regard to their attachments to the inner walls of the frontal sinus and relationship to the frontal recess as anterior or posterior, medial or lateral (Lund et al 2014).

Vascular supply, lymphatic drainage and innervation

The frontal sinuses receive their arterial supply from the supraorbital and anterior ethmoidal arteries. The veins drain into an anastomotic vein in the supraorbital notch that connects the supraorbital and superior ophthalmic veins. Lymphatic drainage is to the submandibular nodes. The sinuses are innervated by branches of the supraorbital nerves (general sensation) and the orbital branches of the pterygopalatine ganglia (parasympathetic secretomotor fibres).

SPHENOIDAL SINUS

The sphenoidal sinuses are two large, irregular cavities within the body of the sphenoid and therefore lie posterior to the upper part of the nasal cavity (see Figs 33.14–33.15; Figs 33.17–33.18). Each opens into the

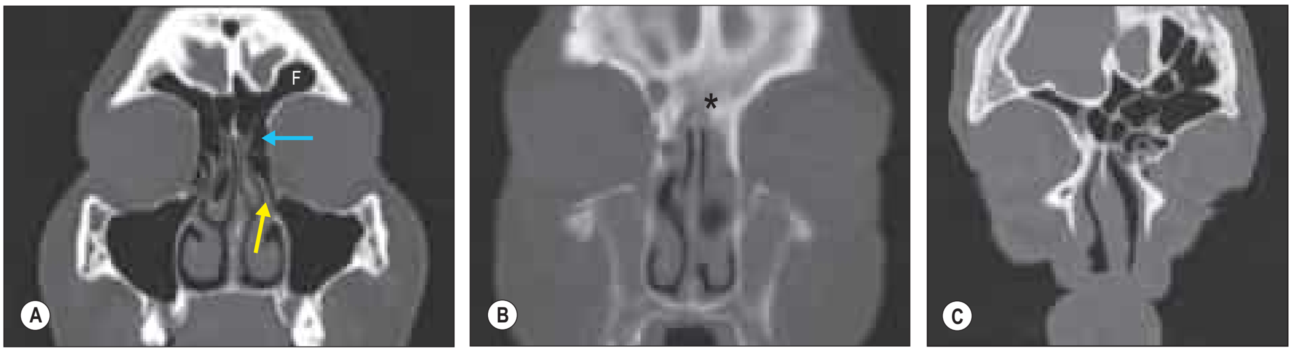


Fig. 33.16 **A**, A coronal CT demonstrating the frontal sinus (F), agger nasi cell (blue arrow) and uncinate process inserting into the middle turbinate (yellow arrow). **B**, An absent frontal sinus (asterisk). **C**, A coronal CT demonstrating variations in the frontal sinus with multiple cells within the frontal recess.

corresponding sphenoethmoidal recess via an aperture high on the anterior wall of the sinus. The sphenoid ostium is usually medial to the superior turbinate, although the height of the ostium is highly variable. The average dimensions of the adult sphenoid are: vertical height 2 cm; transverse breadth 1.8 cm; and anteroposterior depth 2.1 cm. The sinuses are usually separated by a septum that usually deviates from the midline, so that they are unequal in size and form. Their lumina may be further partially divided by bony laminae and accessory septa, especially in the region of former synchondroses, and these septa commonly insert on to the internal carotid artery. Occasionally, one sinus overlaps the other above and, rarely, they intercommunicate. Bony ridges, produced by the internal carotid artery, pterygoid canal, maxillary branch of trigeminal and sometimes the optic nerve, may project into the sinuses from their lateral walls. The sphenoidal sinuses are related above to the optic chiasma and hypophysis cerebri, and on each side to the internal carotid artery and cavernous sinus. One or both sinuses may partially encircle the optic canal. Dehiscences in the osseous walls may occasionally leave their mucosa in contact with the overlying dura mater, optic nerve or carotid artery.

The extent of pneumatization of surrounding bone is highly variable. Sometimes, a lateral recess may extend into the greater and lesser wings of the sphenoid or the pterygoid processes, separating the pterygoid (Vidian) canal and foramen rotundum, and may even invade the basilar part of the occipital bone almost to the foramen magnum. A posterior ethmoidal sinus may extend posterosuperior to the relatively smaller sphenoidal sinuses. In such cases, the sphenoid is medial and inferior to this sphenoethmoidal cell (Onodi cell), which itself will be closely related to the optic nerve and carotid artery. An attempt to approach the sphenoid through a sphenoethmoidal cell places these structures at risk of injury. Sphenoethmoidal cells are readily identified on computed tomographic (CT) imaging, where a horizontal bar can be seen in the coronal plane. The anterior clinoid process may be pneumatized in up to 15% of sphenoidal cells.

The shape and position of the sphenoidal sinus are of clinical importance in an endoscopic trans-sphenoidal surgical approach to the hypophysis cerebri. The sinuses may be classified into three main types: sellar, the most common type, in which the sinus extends for a variable distance beyond the tuberculum sellae; presellar, in which the sinus occasionally extends posteriorly towards, but not beyond, the tuberculum sellae; and conchal, the rarest type, in which a small sinus is separated from the sella turcica by approximately 10 mm of trabecular bone. The anterior midline septum often becomes deviated to one side posteriorly; identification of this septation is important prior to surgery (**Fig. 33.19**).

Vascular supply, lymphatic drainage and innervation

The sphenoidal sinuses receive their arterial supply from the posterior ethmoidal branches of the ophthalmic arteries and nasal branches of the sphenopalatine arteries. Venous drainage is through the posterior ethmoidal veins draining into the superior ophthalmic veins. Lymph drainage is to the retropharyngeal nodes. The sinuses are innervated by the posterior ethmoidal branches of the ophthalmic nerves (general sensation) and the orbital branches of the pterygopalatine ganglia (parasympathetic secretomotor fibres).

ETHMOIDAL SINUSES

The ethmoidal sinuses differ from the other paranasal sinuses in that they are formed of multiple thin-walled cavities in the ethmoidal labyrinth (see Figs 33.13, 33.15–33.18). The number and size of the cavities vary, from 3 large to 18 small sinuses on each side. They lie between the upper part of the nasal cavity and the orbit, and are separated from the latter by the paper-thin lamina papyracea or orbital plate of the ethmoid (this presents a poor barrier to infection, which may therefore spread into the orbit). Pneumatization may extend into the middle concha, or into the body and wings of the sphenoid bone lateral to the sphenoidal sinus. (There is a view that the ethmoidal sinuses are aerated after birth, rather than pneumatized, as happens with the other paranasal sinuses (Jankowski 2013).)

The ethmoidal sinuses are divided clinically into anterior and posterior groups on each side, distinguished by their embryological development, sites of communication with the nasal cavity, their mucociliary drainage, and their relation to the basal lamella of the middle turbinate. (Cells wrongly designated previously as belonging to a middle group are now included with the anterior group (Stammberger and Kennedy 1995).) The anterior and posterior groups are separated from each other by the basal lamella; this may be indented by cells from either group and therefore it forms a rather tortuous barrier between them. Within each group, the sinuses are only partially separated by incomplete bony septa.

Anterior ethmoidal sinuses

Up to 11 anterior ethmoidal air cells drain into the ethmoidal infundibulum, a three-dimensional, funnel-shaped cleft between the uncinate and lateral wall of the nose, by one or more orifices. The most anterior group, developmental remnants of the first ethmoturbinals, are the agger nasi cells. These cells are almost always present. They are medial relations of the lacrimal sac and duct, and invaginate beneath the frontal sinus on the lateral wall of the nasal cavity anteriorly. Removal of the superior walls of the agger nasi, 'uncapping the egg', is a key part of surgery of the frontal recess. The ethmoidal bulla, arising from the second ethmoturbinal of the nose, consists of a group of the largest and least variable anterior ethmoidal cells. The cleft between the posterior edge of the uncinate bone and the face of the ethmoidal bulla is known as the hiatus semilunaris. When the bullar lamella reaches the cranial base, this forms the posterior wall of the frontal recess. In other cases, a suprabullar recess may be found. Infraorbital (Haller) cells may develop medially beneath the orbital floor (**Fig. 33.20**).

Posterior ethmoidal sinuses

Up to seven posterior ethmoidal air cells usually drain by a single orifice into the superior meatus; one may drain into the supreme meatus when present, and one or more into the sphenoidal sinus (**Fig. 33.21**). The posterior group lies very close to the optic canal and optic nerve; optic nerve injury is a devastating potential complication of endoscopic sinus surgery, particularly if a sphenoethmoidal cell (Onodi cell) is present. The reported incidence varies widely (3.5 to 51%), according to racial group; it is more common in non-Caucasians. The sphenoethmoidal cell is usually regarded as the most posterior ethmoidal cell that pneumatizes lateral and superior to the sphenoidal sinus, and is intimately associated with the optic nerve; it may contain a tuberculum nervi optici, where the optic canal bulges into the wall of the cell.

Roof of the ethmoid

The ethmoid bone is open superiorly; the roof is closed by the orbital plate of the frontal bone. Ethmoidal air cells indent this plate; each one is a 'fovea ethmoidalis'. The thin, lateral cribriform lamella (one of the thinnest parts of the cranial base) forms the medial wall of the roof, extending from the middle turbinate to the cribriform plate, and the lateral wall of the olfactory fossa or niche. The olfactory fossa varies in depth and is frequently asymmetrical; it is at risk during sinus surgery.

Uncinate process

The uncinate process in a thin, hook-shaped bone that articulates with the perpendicular plate of the palatine bone and the ethmoidal process of the inferior concha. Its posterosuperior margin is free, forming the hiatus semilunaris with the ethmoidal bulla. It may sometimes reflect medially into the middle meatus and usually overlies the ostium of the maxillary sinus. It is frequently removed during sinus surgery (Becker 1994).

Vascular supply and innervation

The ethmoidal sinuses receive their arterial supply from nasal branches of the sphenopalatine artery and the anterior and posterior ethmoidal branches of the ophthalmic artery. Venous drainage is by the corresponding veins. The lymphatics of the anterior group drain to the submandibular nodes, and those of the posterior group to the retropharyngeal nodes. The sinuses are innervated by the anterior and posterior ethmoidal branches of the ophthalmic nerves (general sensation) and the orbital branches of the pterygopalatine ganglia (parasympathetic secretomotor fibres).

MAXILLARY SINUS

The maxillary sinus is the largest of the paranasal sinuses. It usually fills the body of the maxilla and is pyramidal in shape (see Figs 30.8B, 33.13, 33.17; **Fig. 33.22**). The base is medial and forms much of the lateral wall of the nasal cavity. The floor, which often lies below the nasal floor, is formed by the alveolar process and part of the palatine process of the maxilla. It is related to the roots of the teeth, especially the second premolar and first molar, but may extend posteriorly to the third molar tooth and/or anteriorly to incorporate the first premolar, and sometimes the canine. Defects in the bone overlying the roots are

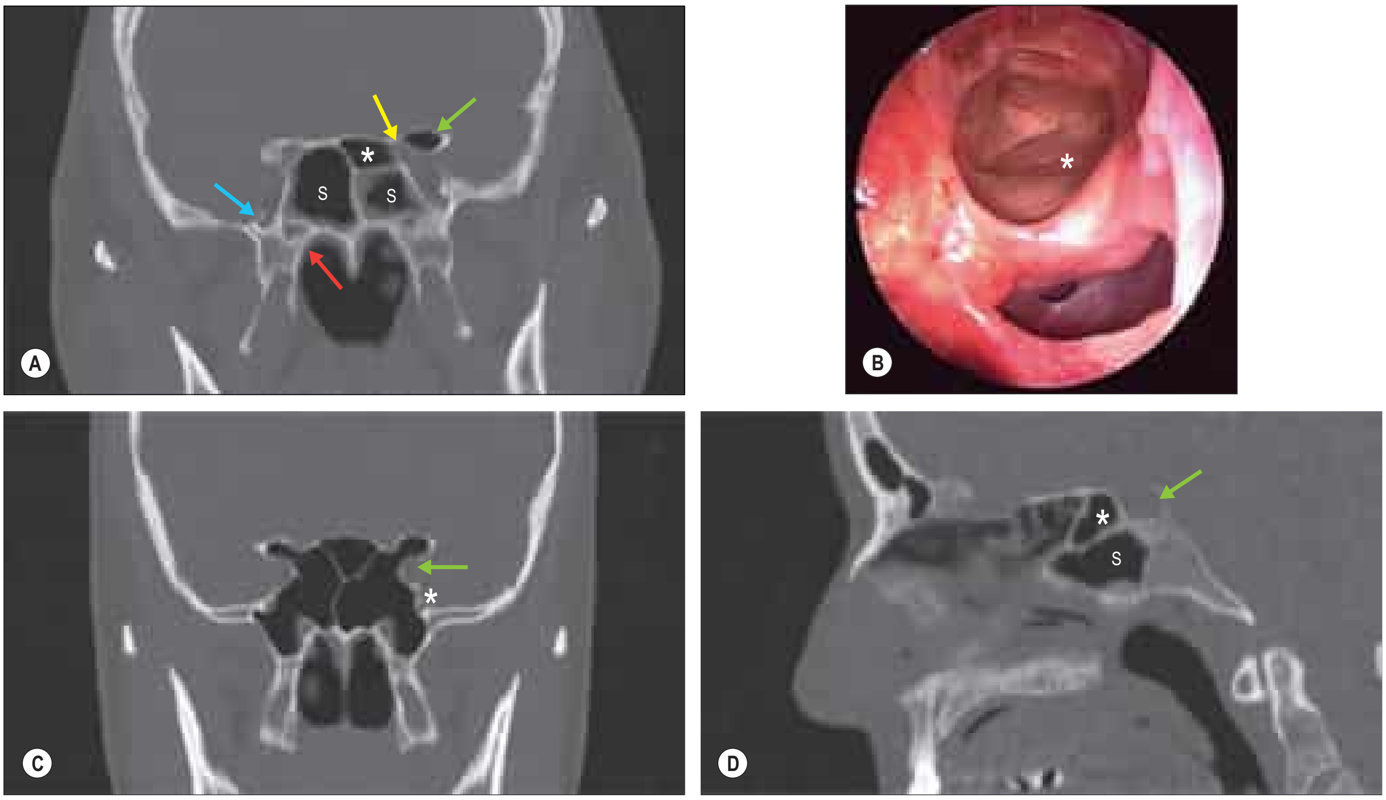


Fig. 33.19 **A**, A coronal CT demonstrating a normal right-sided sphenoidal sinus and a small left-sided sphenoidal sinus (S). The left sphenoethmoidal cell is also shown (asterisk). Pneumatization of the left anterior clinoid process (green arrow) means that the optic nerve (yellow arrow) lies exposed within the sphenoethmoidal cell. The foramen rotundum (blue arrow) and pterygoid (Vidian) canal (red arrow) can also be seen. **B**, An endoscopic view of a right Onodi cell; the optic nerve is visible in the posterolateral wall of the cell (asterisk). **C**, A coronal CT demonstrating well-pneumatized sphenoidal sinuses. The foramen rotundum (asterisk) and internal carotid artery (arrow) are also shown. **D**, A sagittal CT demonstrating the sphenoidal sinus (S), pituitary fossa (arrow) and posterior sphenoethmoidal cell (asterisk).

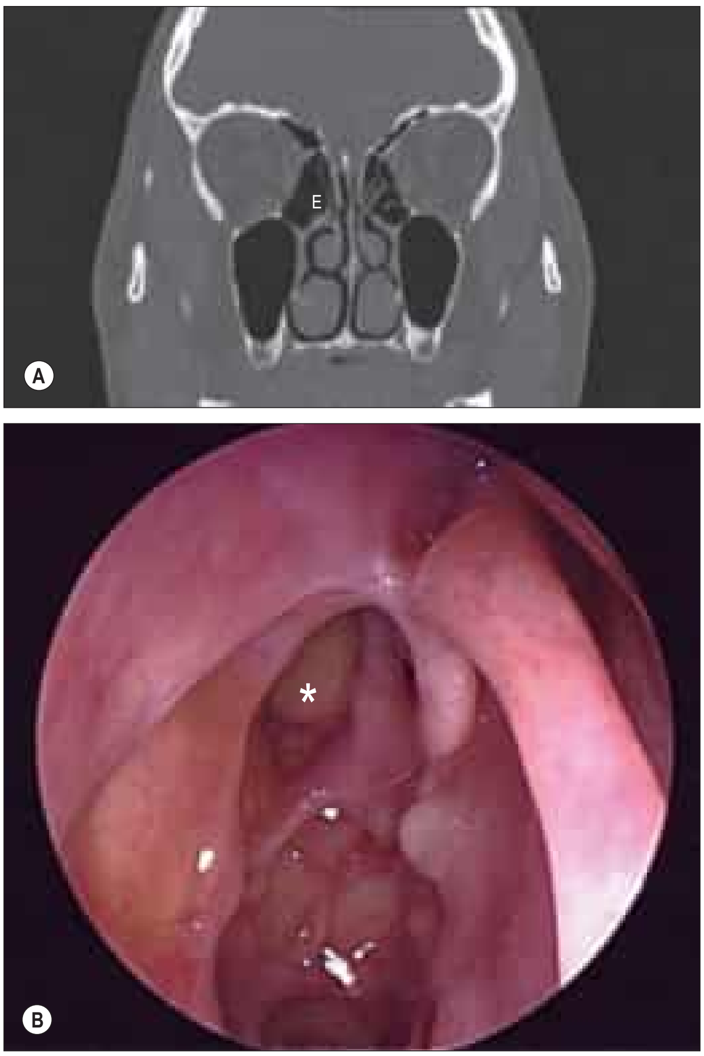


Fig. 33.20 **A**, The anterior ethmoidal air cells (E). **B**, An endoscopic view showing a partially opened agger nasi cell (asterisk). **C**, An infraorbital ethmoidal cell (Haller cell, asterisk), seen as a variation of anterior ethmoidal anatomy.

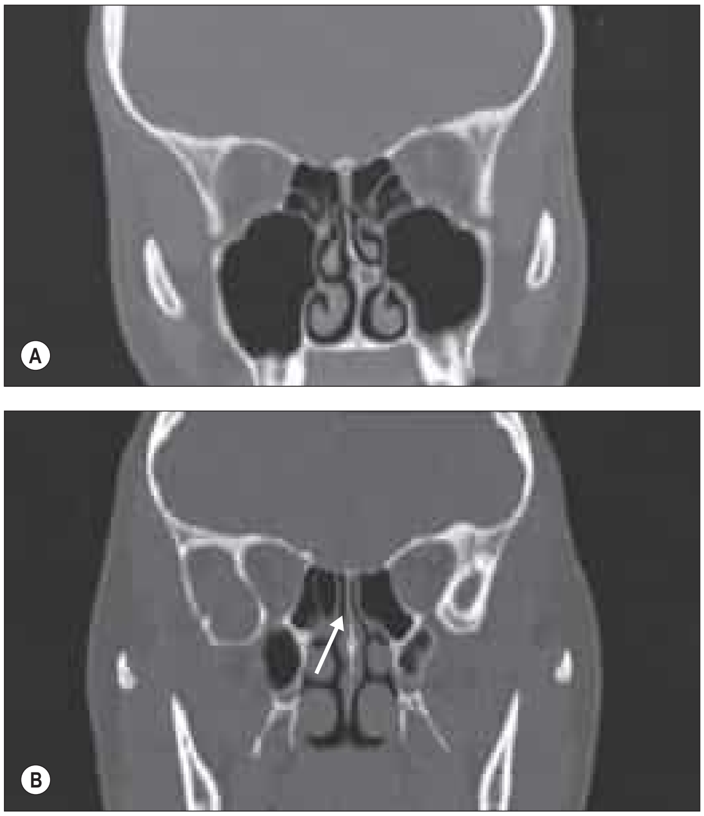


Fig. 33.21 **A**, A coronal CT demonstrating the posterior ethmoidal cells. A bony spur can be seen in the nasal septum. **B**, The posterior ethmoidal air cells and the sphenoethmoidal recess (arrow).

not uncommon. The roof of the sinus forms the major part of the floor of the orbit. It contains the infraorbital canal, which may exhibit dehiscences. The lateral truncated apex of the pyramid extends into the zygomatic process of the maxilla, and may reach the zygomatic bone, in which case it forms the zygomatic recess, which throws a V-shaped shadow over the antrum on a lateral radiograph.

The facial surface of the maxilla forms its anterior wall, and is grooved internally by a delicate canal (canalis sinuosus) that houses the anterior superior alveolar nerve and vessels as they pass forwards from the infraorbital canal. The posterior wall is formed by the infratemporal surface of the maxilla; it contains alveolar canals that may produce ridges in the sinus and that also conduct the posterior superior alveolar vessels and nerves to the molar teeth. The medial wall is deficient posterosuperiorly at the maxillary hiatus, a large opening that is partially closed in an articulated skull by portions of the perpendicular plate of the palatine bone, the uncinate process of the ethmoid bone, the inferior nasal concha, the lacrimal bone and the overlying nasal mucosa, to form an ostium and anterior and posterior fontanelles. The ostium usually opens into the inferior part of the ethmoidal infundibulum, and thence into the middle meatus, via the hiatus semilunaris (the hiatus forms the area above the superior edge of the uncinate process). The ostium is not normally visible on examination of the nose, without prior resection of the uncinate process. The fontanelles are covered only by periosteum and mucosa, and may contain accessory ostia that may be visible on nasendoscopy and CT. All of the openings are nearer the roof than the floor of the sinus, which means that the natural drainage of the maxillary sinus is reliant on an intact mucociliary escalator; the cilia of the sinus mucoperiosteum normally beat towards the ostium.

The maxillary sinus may be incompletely divided by septa; complete septa are very rare. The thinness of its walls is clinically significant in determining the spread of tumours from the maxillary sinus. A tumour may push up the orbital floor and displace the eyeball; project into the nasal cavity, causing nasal obstruction and bleeding; protrude on to the cheek, causing swelling and numbness if the infraorbital nerve is damaged; spread back into the infratemporal fossa, causing restriction of mouth opening due to pterygoid muscle damage and pain; or spread down into the mouth, loosening teeth and causing malocclusion. Extraction of molar teeth may damage the floor, and impact may fracture its walls. Rarely, hypoplasia of one maxillary antrum is present.

Ostiomeatal complex

The term ostiomeatal complex, or ostiomeatal unit, refers to the area that includes the maxillary sinus ostium, ethmoidal infundibulum and the hiatus semilunaris (see Fig. 33.17); it is a functional complex rather than a clearly defined anatomical structure (see Fig. 33.22A). The complex is the common pathway for drainage of secretions from the maxillary and anterior group of ethmoidal sinuses; where the uncinate process attaches to the lateral nasal wall, the complex also drains the frontal sinus.

Vascular supply, lymphatic drainage and innervation

The arterial supply of the maxilla is derived mainly from the maxillary arteries via the anterior, middle and posterior superior alveolar branches and from the infraorbital and greater palatine arteries. Branches of the posterior superior alveolar artery and the infraorbital artery form an anastomosis in the bony wall of the sinus, which also supplies the mucous membrane that lines the nasal chambers. An extraosseous anastomosis frequently exists between the posterior superior alveolar artery and the infraorbital artery. The intra- and extraosseous anastomoses form a double arterial arcade that supplies the lateral antral wall and, partly, the alveolar process. Veins corresponding to the arteries drain into the facial vein or pterygoid venous plexus on either side. Lymph drainage is to the submandibular nodes. The sinuses are innervated by the infraorbital and anterior, middle and posterior superior alveolar branches of the maxillary nerves (general sensation), and nasal branches of the pterygopalatine ganglia (parasympathetic secretomotor fibres).

IMAGING OF THE PARANASAL SINUSES

Standard radiological images are no longer recommended in the diagnosis of rhinosinusitis because of their poor specificity and sensitivity, although they may be used to confirm the presence of acute frontal or maxillary sinusitis that has failed to respond to medical treatment and requires urgent drainage. CT defines anatomical variations but should not be used in isolation for diagnosis because 1 in 3 asymptomatic individuals show incidental mucosal changes. Spiral CT provides good-quality axial, coronal and sagittal images that facilitate an appreciation of the size and relationship of the paranasal sinuses (**Fig. 33.23**).

SPREAD OF INFECTION FROM THE SINUSES

Uncontrolled paranasal sinus infection can cause very significant morbidity. In the pre-antibiotic era, it was often associated with mortality from meningitis and brain abscess. Paranasal sinus infection has the potential to spread to the orbit, cavernous sinuses, meninges and brain. The ability to overcome infection at this site depends on the virulence of the infecting organism, the speed with which appropriate treatment is delivered, innate immunity and individual anatomical aspects of the sinuses that may predispose to spread of infection. Normal mucociliary clearance of the nasal and paranasal mucosa becomes paralysed or uncoordinated very quickly with the onset of infection and patent or potentially patent drainage pathways become paramount. The middle meatus forms the common drainage pathway for the anterior ethmoidal, frontal and maxillary sinuses. The posterior ethmoidal and sphenoidal sinuses drain into the superior meatus and sphenoethmoidal recess. Endoscopic examination will usually show infected mucus draining from these areas in this situation (Simmen and Jones 2005).

The bony walls of the sinuses are paper-thin in places and dehiscences of them, particularly of the lamina papyracea and cribriform plate of the ethmoids, the lateral wall of the sphenoid, and the orbital and posterior walls of the frontal sinus, bring infected sinus mucosa into direct contact with orbital periosteum, the dura of the anterior cranial fossa and the cavernous sinus. Septic thrombophlebitis then develops and infection spreads rapidly by this route. Sequelae can include blindness, intra- and extradural collections, cavernous sinus thrombosis, meningitis, frontal lobe abscess and osteomyelitis of the cranial vault if diploic veins are involved.

Bonus e-book images and table

Fig. 33.2 The adult male nose. **A**, Basal view. **B**, Frontal view.

Fig. 33.7 A, An endoscopic view of the nasal cavity showing inferior and middle turbinates. **B**, A view of the middle meatus and left lateral wall of the nose. **C**, An endoscopic view of the nose showing the middle turbinate and the pneumatized uncinate process in the middle meatus. **D**, An endoscopic view of the nose demonstrating a deviated nasal septum making contact with the left inferior turbinate. **E**, An endoscopic view of the posterior nasal space.

Fig. 33.8 C, A higher-power view of the expanded end of an olfactory receptor neurone. **D**, A section of human olfactory epithelium immunostained with anti-OMP (olfactory marker protein).

Fig. 33.16 A, A coronal CT demonstrating the frontal sinus, agger nasi cell and uncinate process inserting into the middle turbinate. **B**, An absent frontal sinus. **C**, A coronal CT demonstrating variations in the frontal sinus with multiple cells within the frontal recess.

Fig. 33.19 A, A coronal CT demonstrating a normal right-sided sphenoidal sinus and a small left-sided sphenoidal sinus. **B**, An endoscopic view of a right Onodi cell; the optic nerve is visible in the posterolateral wall of the cell. **C**, A coronal CT demonstrating well-pneumatized sphenoidal sinuses. **D**, A sagittal CT demonstrating the sphenoidal sinus, pituitary fossa and posterior sphenoethmoidal cell.

Fig. 33.20 A, The anterior ethmoidal air cells. **B**, An endoscopic view showing a partially opened agger nasi cell. **C**, An

Continued

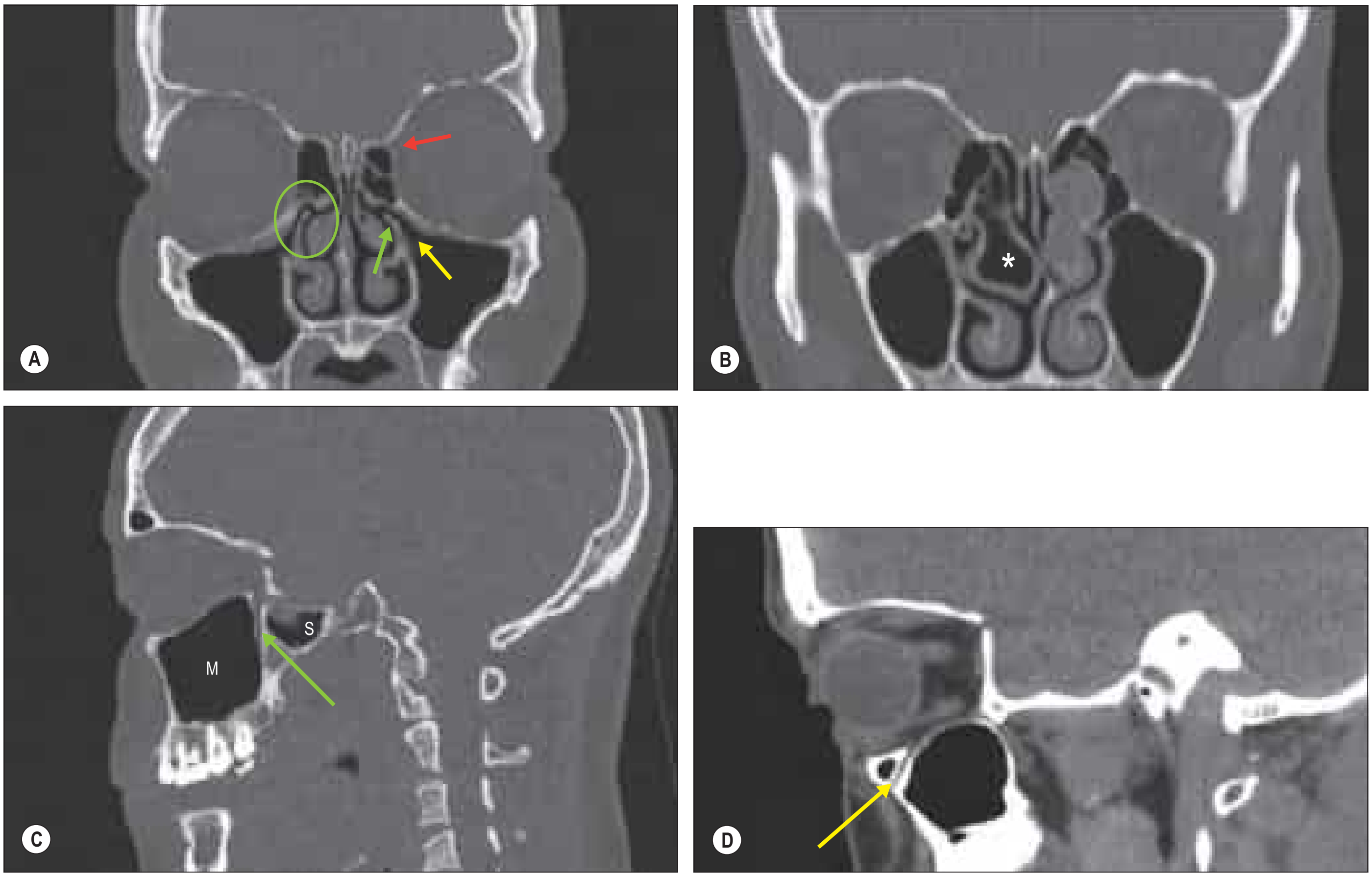


Fig. 33.22 **A**, A coronal section demonstrating the maxillary and anterior ethmoidal sinuses. The ostiomeatal complex (ringed) and infundibulum (yellow arrow) are shown. The anterior ethmoidal artery can be seen leaving the orbit (red arrow). The uncinate process is also visible (green arrow). **B**, A coronal CT demonstrating the maxillary sinuses and a right-sided, pneumatized middle concha (concha bullosa; asterisk). **C**, A sagittal CT demonstrating the relationship between the posterior wall of the maxillary sinus (M), the pterygopalatine fossa (arrow) and a lateral extension of the sphenoidal sinus (S). **D**, A sagittal CT showing the course of the infraorbital nerve through the maxillary sinus (arrow).

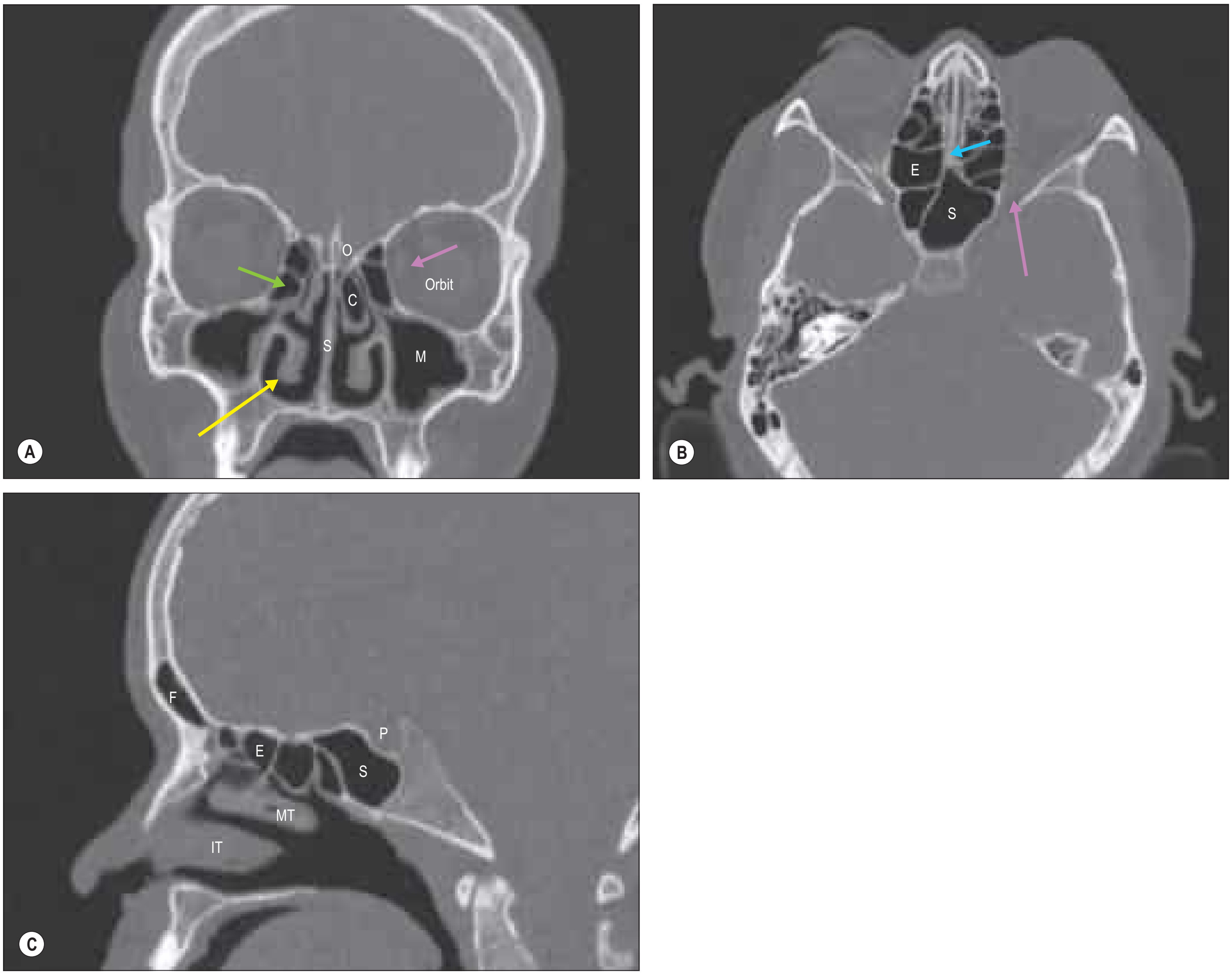


Fig. 33.23 **A**, A coronal section of the skull. Medial rectus (purple arrow); inferior turbinate (yellow arrow); anterior ethmoidal air cells (green arrow). Abbreviations: C, concha bullosa (pneumatized middle turbinate); M, maxillary sinus; O, olfactory cleft; S, nasal septum. **B**, An axial section of the skull. Medial rectus (blue arrow); orbital apex (purple arrow). Abbreviations: E, ethmoidal sinus; S, sphenoidal sinus. **C**, A sagittal section of the skull. Abbreviations: E, ethmoidal sinus; F, frontal sinus; IT, inferior turbinate; MT, middle turbinate; P, pituitary fossa; S, sphenoidal sinus.

Bonus e-book images and table—cont'd

infraorbital ethmoidal cell (Haller cell), seen as a variation of anterior ethmoidal anatomy.

Fig. 33.21 A, A coronal CT demonstrating the posterior ethmoidal cells. **B**, The posterior ethmoidal air cells and the sphenoethmoidal recess.

Fig. 33.22 A, A coronal section demonstrating the maxillary and anterior ethmoidal sinuses. **B**, A coronal CT demonstrating the maxillary sinuses and a right-sided, pneumatized middle concha (concha bullosa). **C**, A sagittal CT demonstrating the relationship between the posterior wall of the maxillary sinus, the pterygopalatine fossa and a lateral extension of the sphenoidal sinus. **D**, A sagittal CT showing the course of the infraorbital nerve through the maxillary sinus.

Fig. 33.23 A, A coronal section of the skull. **B**, An axial section of the skull. **C**, A sagittal section of the skull.

Table 33.1 Terminology.

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